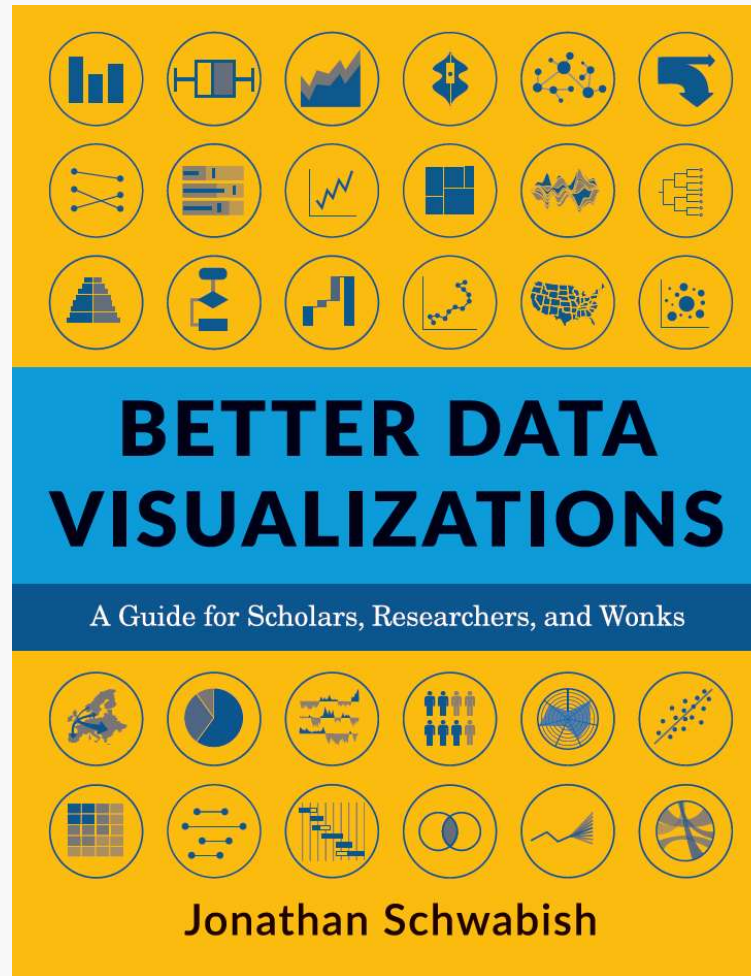


DSCE Practical 7

Data visualization principles

Riccardo Caraccio

Inspiration

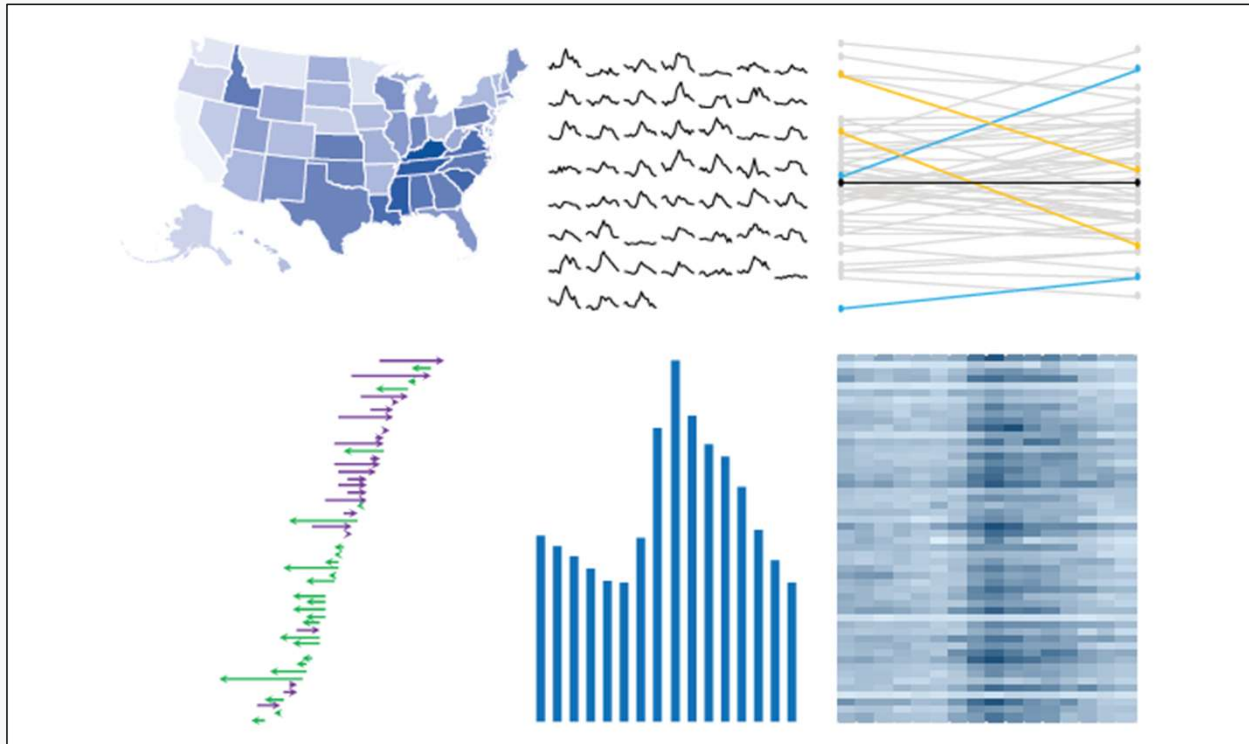


Outline

- **Why data visualization is important**
- **General principle of Data Visualization**
- **In depth on some plots:**
 - **Bar plot**
 - **Time series**
 - **Tables**
- **Try it yourself**
- **Showcase**

Why data visualization

Why data visualization



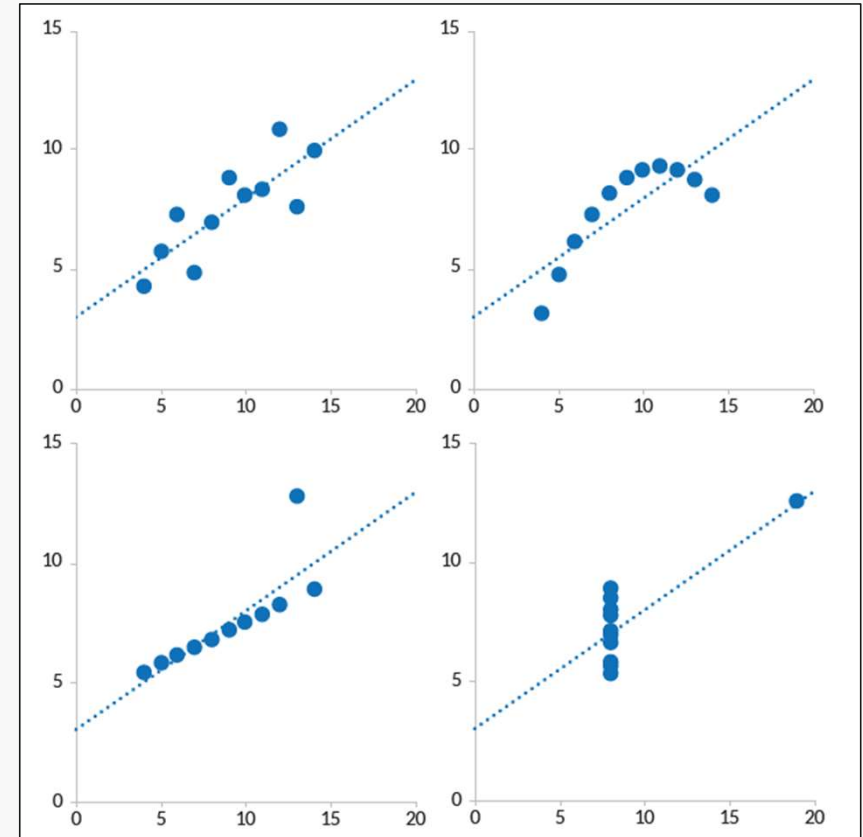
Each of these six charts visualizes the same data: The share of people earning minimum wage or less in each state.

Why data visualization

Anscombe's Quartet

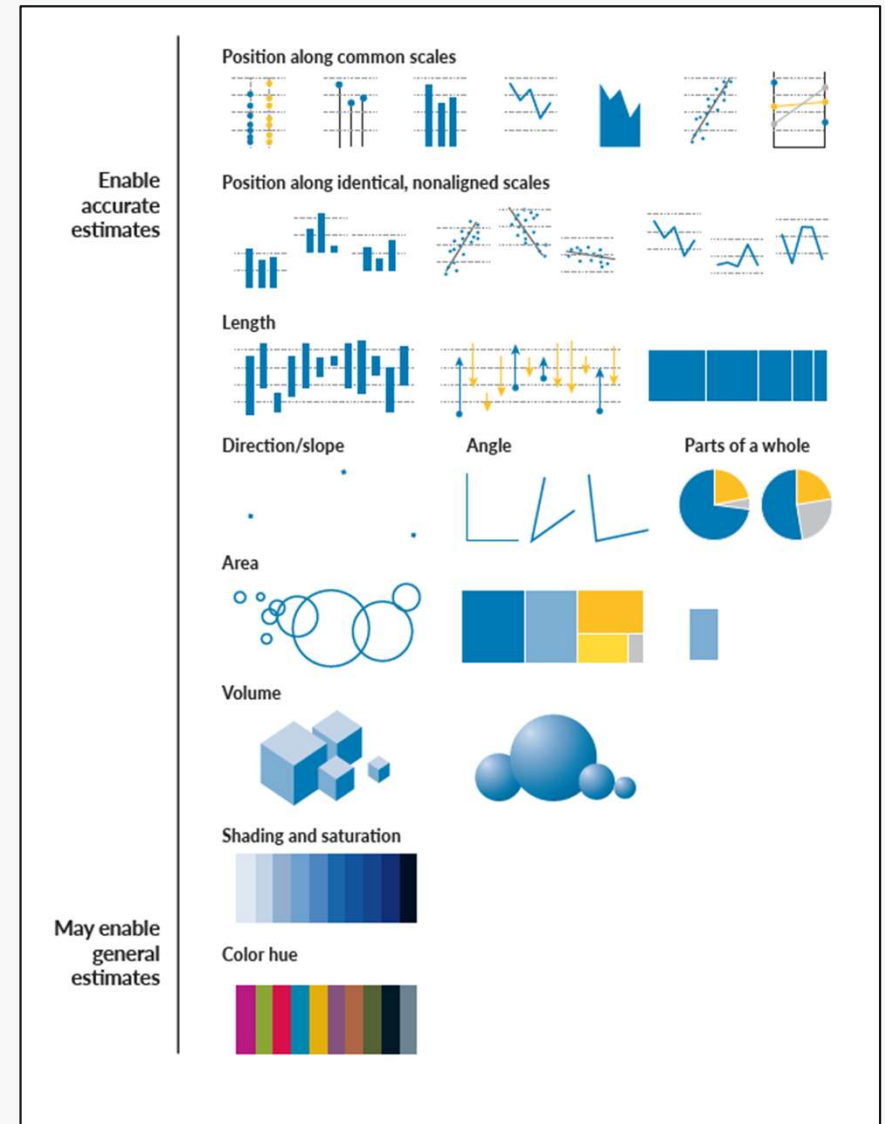
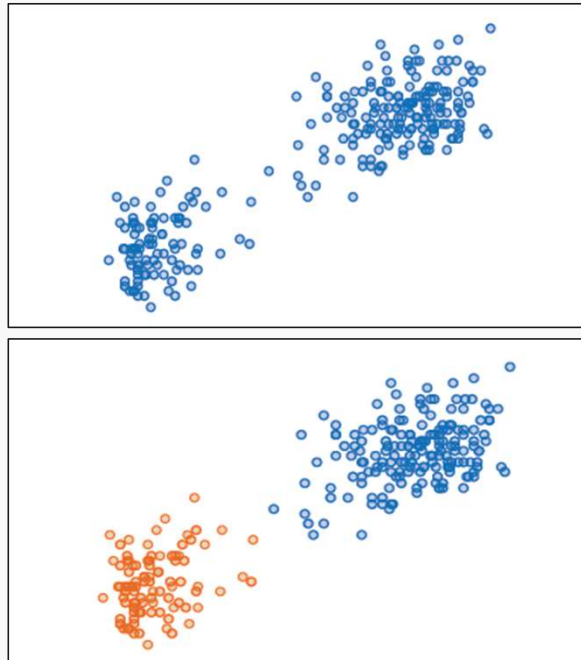
Data set		1	1	2	2	3	3	4	4
Variable		x	y	x	y	x	y	x	y
Obs. No.	1 :	10	8.0	10	9.1	10	7.5	8	6.6
	2 :	8	7.0	8	8.1	8	6.8	8	5.8
	3 :	13	7.6	13	8.7	13	12.7	8	7.7
	4 :	9	8.8	9	8.8	9	7.1	8	8.8
	5 :	11	8.3	11	9.3	11	7.8	8	8.5
	6 :	14	10.0	14	8.1	14	8.8	8	7.0
	7 :	6	7.2	6	6.1	6	6.1	8	5.3
	8 :	4	4.3	4	3.1	4	5.4	19	12.5
	9 :	12	10.8	12	9.1	12	8.2	8	5.6
	10 :	7	4.8	7	7.3	7	6.4	8	7.9
	11 :	5	5.7	5	4.7	5	5.7	8	6.9
Mean		9.0	7.5	9.0	7.5	9.0	7.5	9.0	7.5
Variance		11.0	4.1	11.0	4.1	11.0	4.1	11.0	4.1
Correlation		0.816		0.816		0.816		0.817	
Regression line		$y = 3 + 0.5x$		$y = 3 + 0.5x$		$y = 3 + 0.5x$		$y = 3 + 0.5x$	

Source: Francis Anscombe

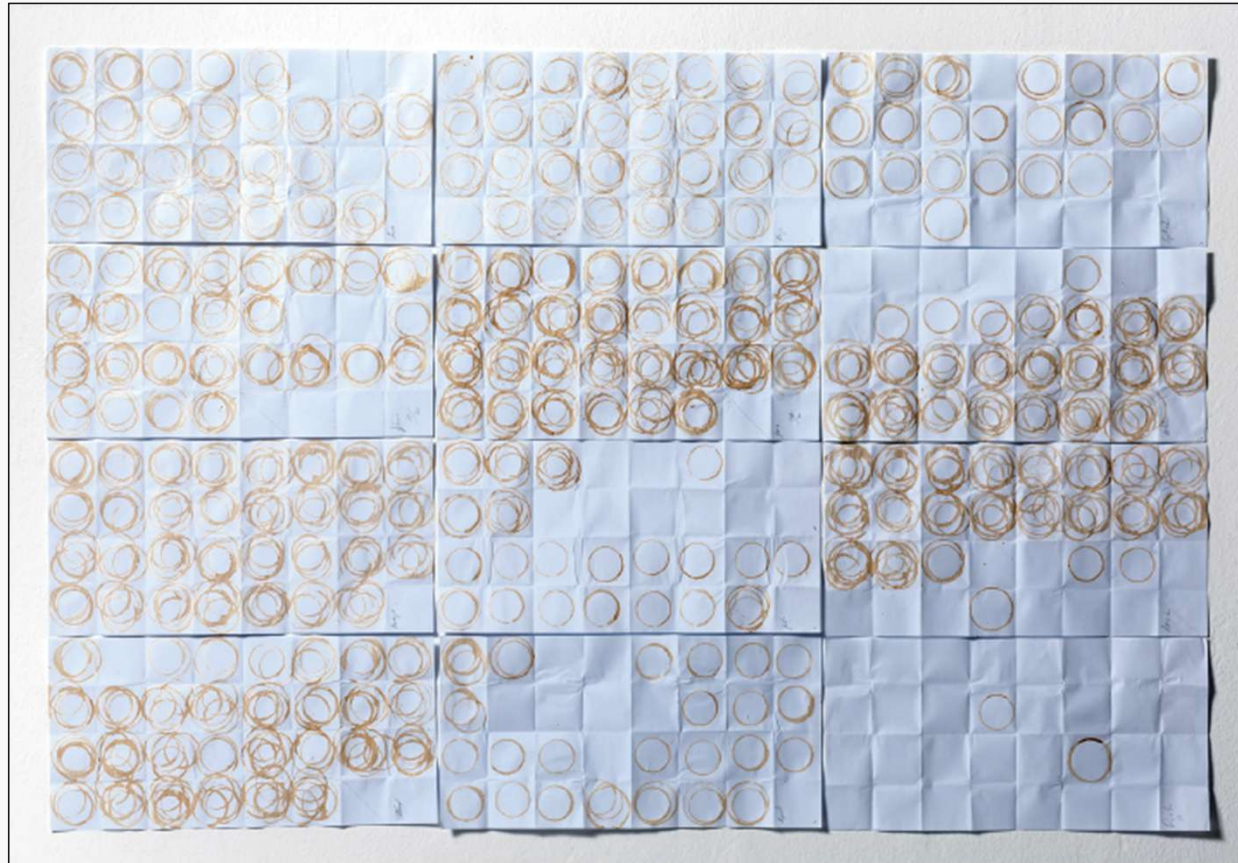


Why data visualization

How does the reader perceive our data?



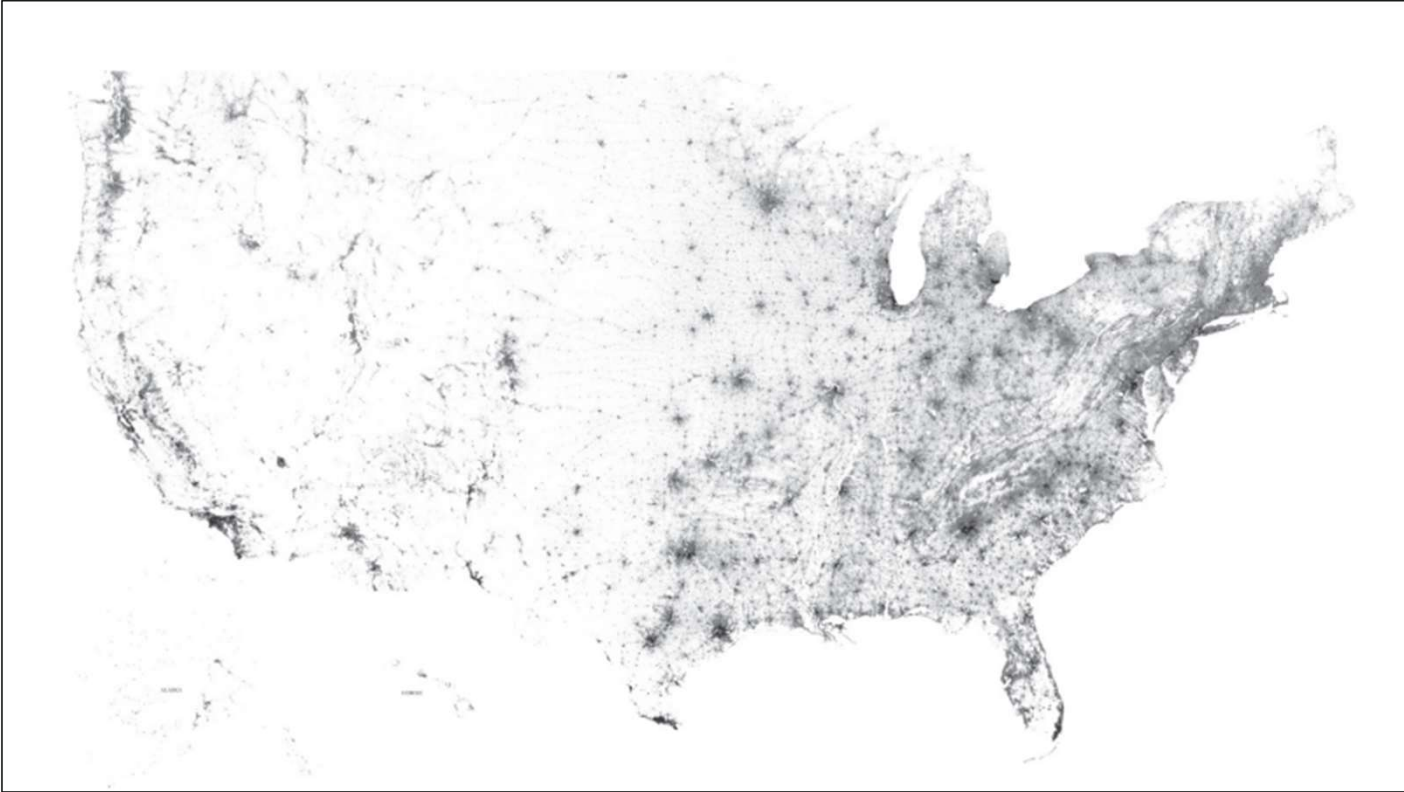
Don't make the data boring



Artist and journalist Jaime Serra Palou plotted his coffee consumption every day for a year by using stains from his coffee cup.

General principles of data visualization

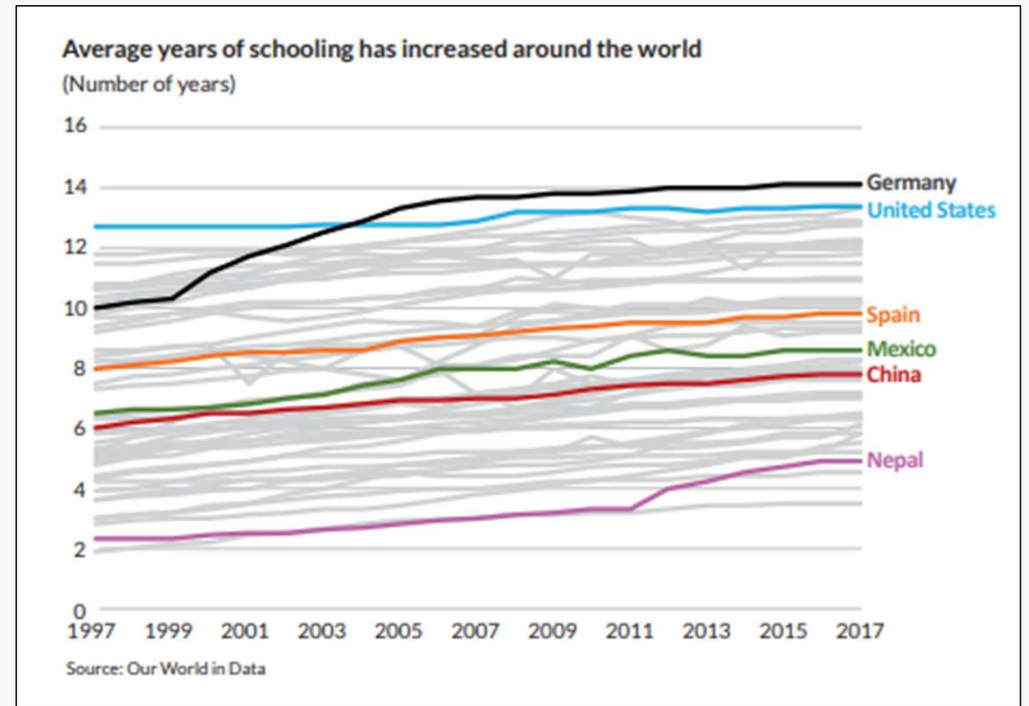
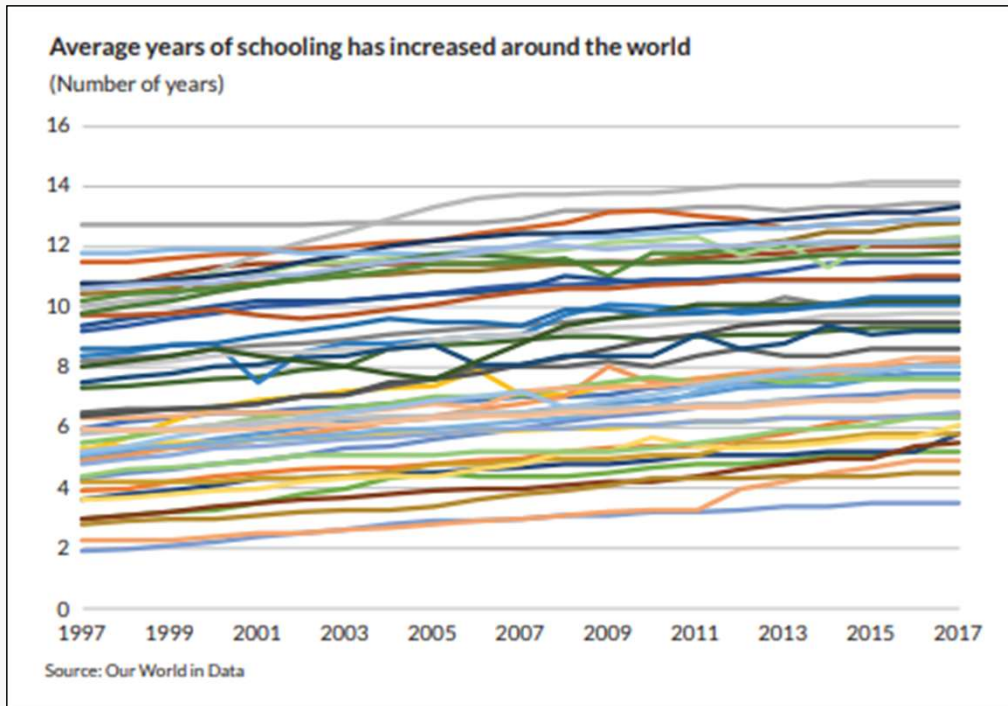
Show the data!



Dot density map of the United States. It uses data from the 2010 U.S. decennial census and places a dot for each of the country's 308 million residents in their census blocks (a census block roughly corresponds to a city block).

Image Copyright, 2013, Weldon Cooper Center for Public Service, Rector and Visitors of the University of Virginia (Dustin A. Cable, creator).

Show the data... but not all the data is necessary



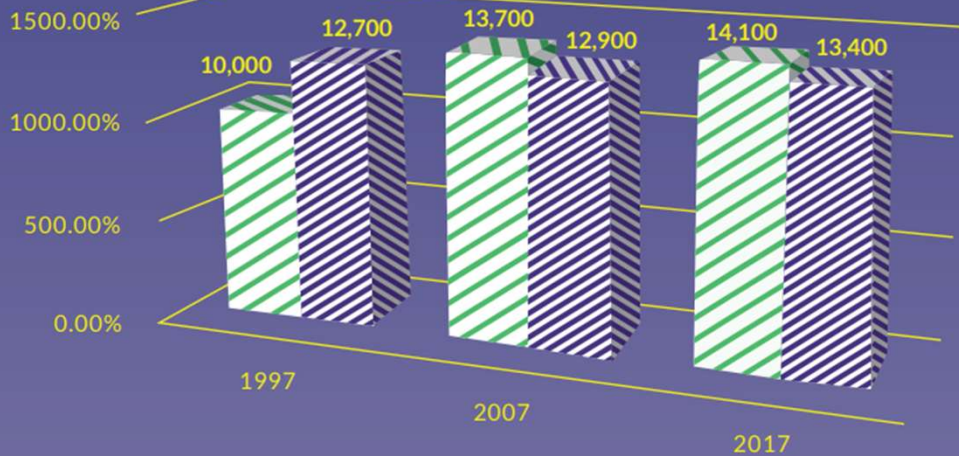
Highlighting just a few countries in the chart on the right makes it easier to read and make your point

Reduce the clutter

Average years of schooling has grown faster in Germany than in the United States

(Number of years)

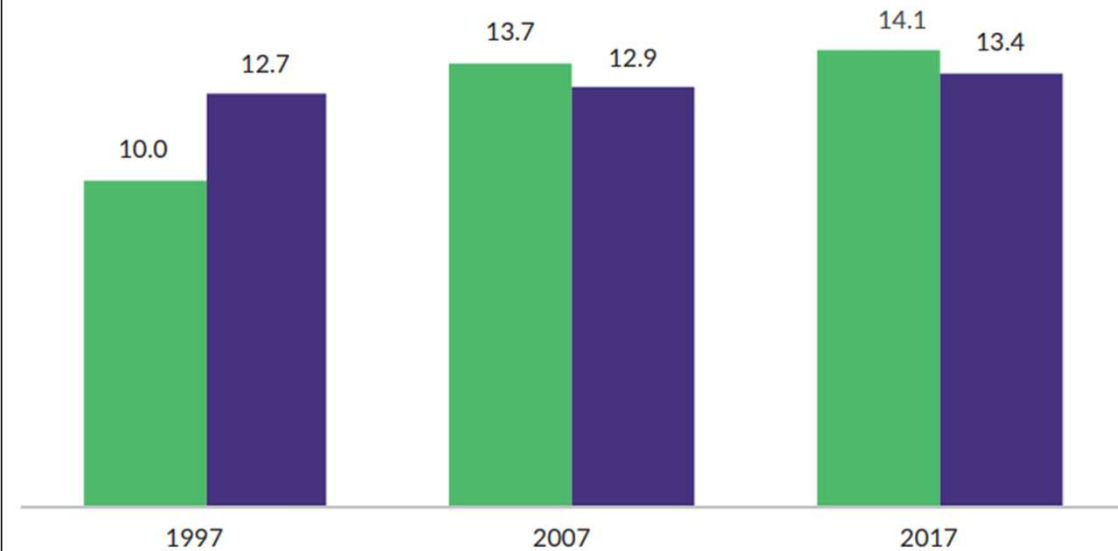
Germany United States



Average years of schooling has grown faster in Germany than in the United States

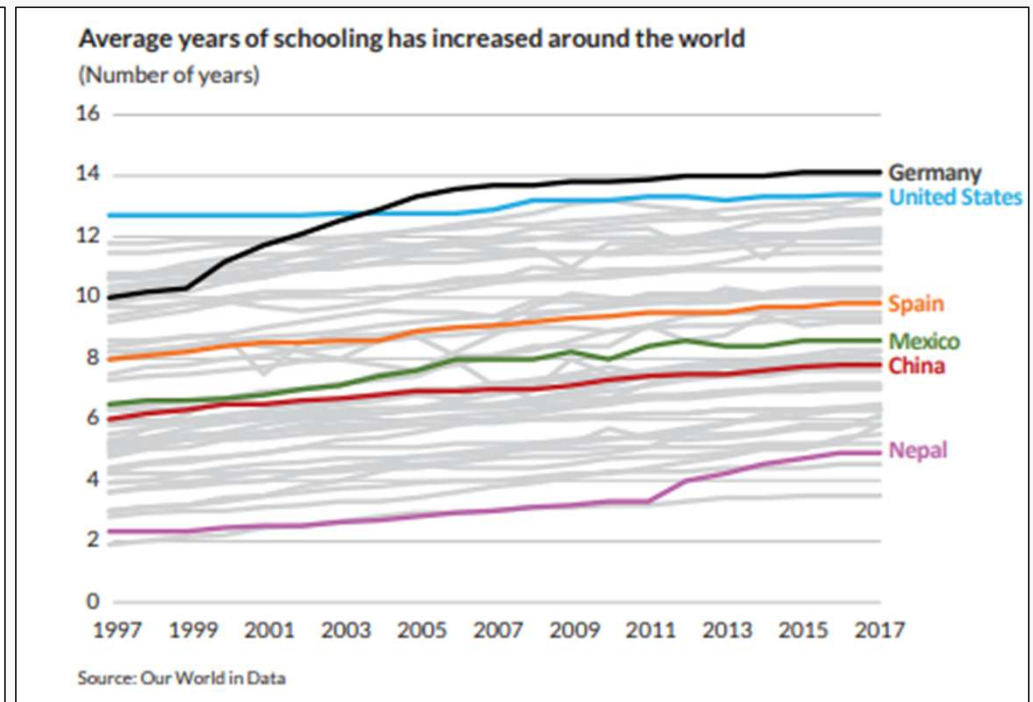
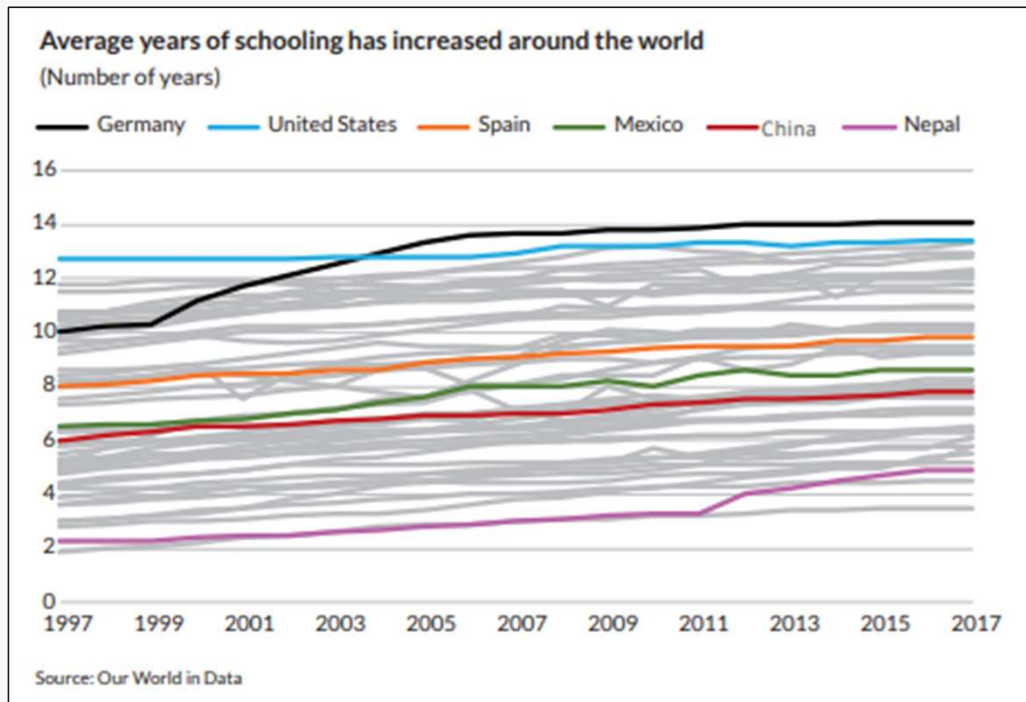
(Number of years)

Germany United States



Integrate the graphics and text

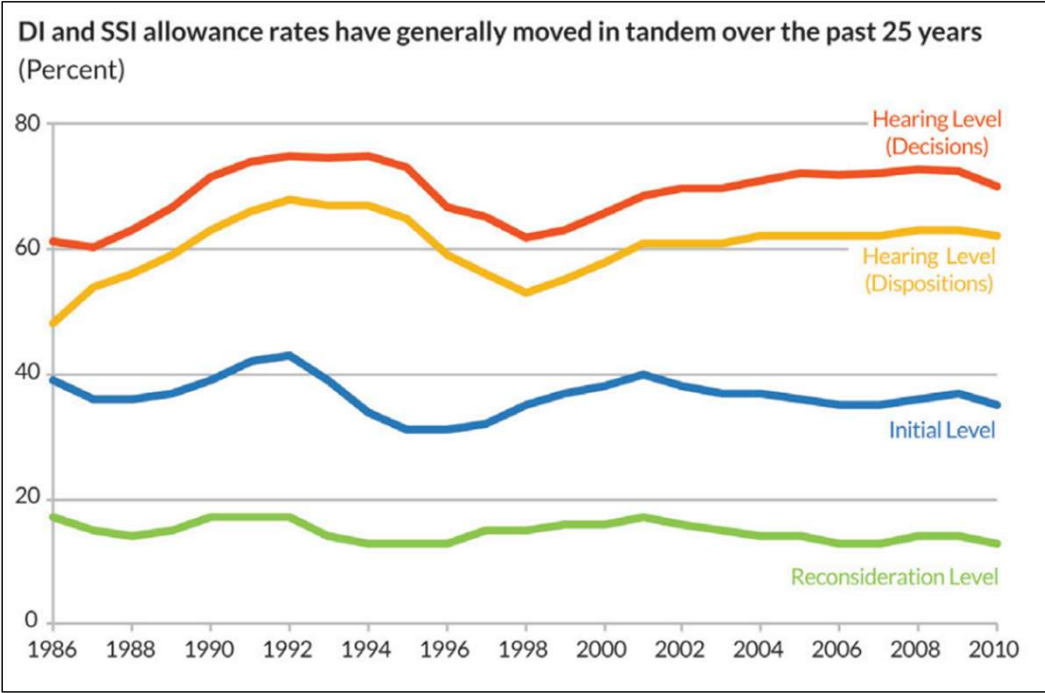
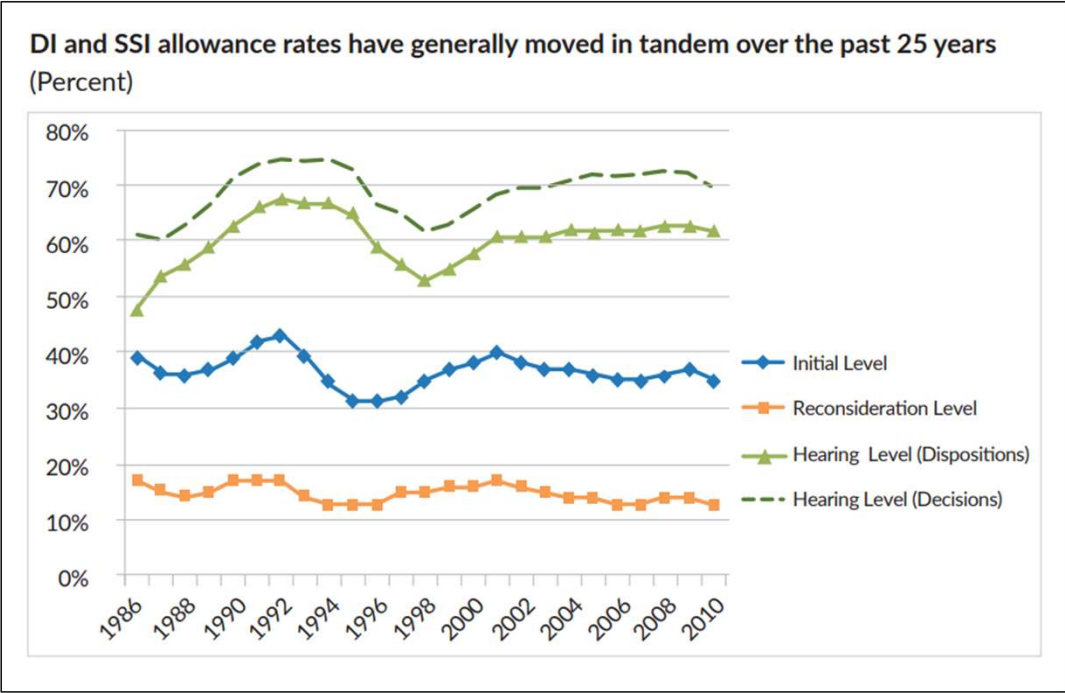
Remove legends when possible and label data directly



Also, align the labels and match the colors with the data as in the graph on the right rather than placing them in random positions.

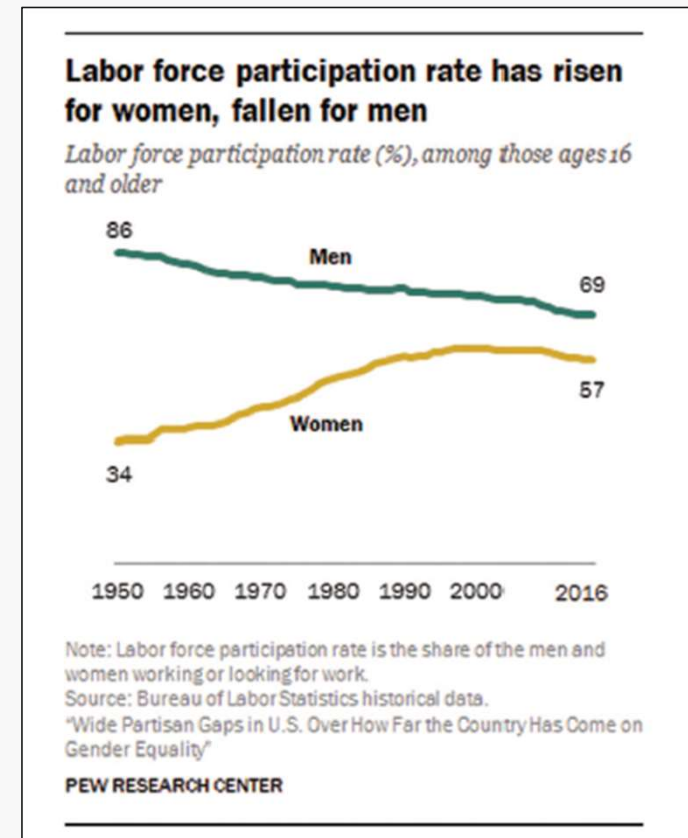
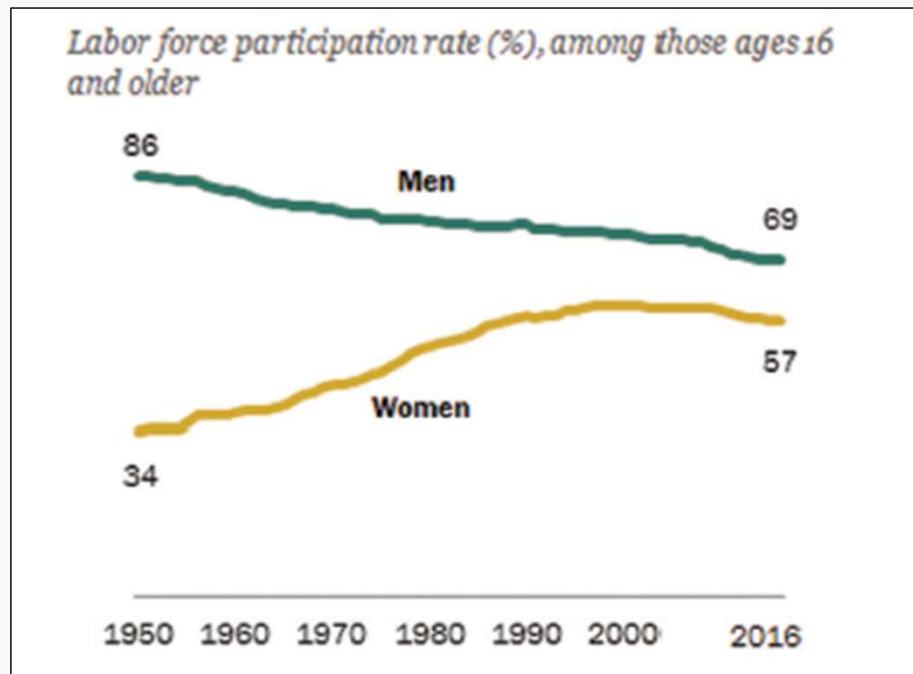
Integrate the graphics and text

Remove legends when possible and label data directly



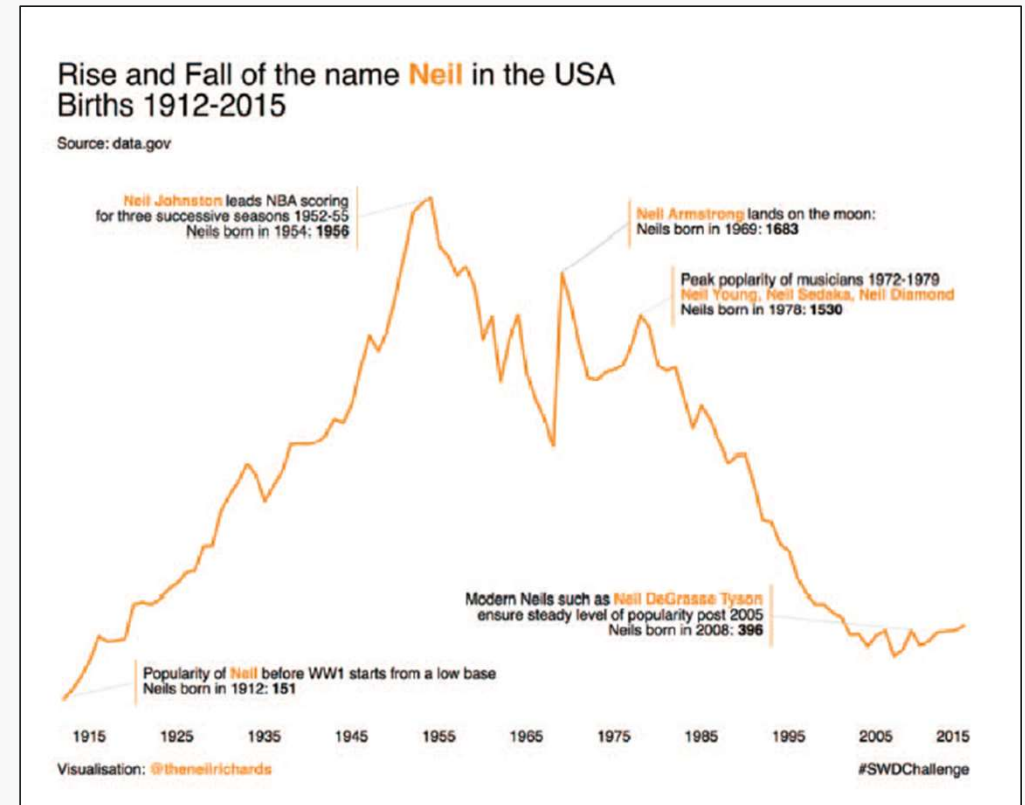
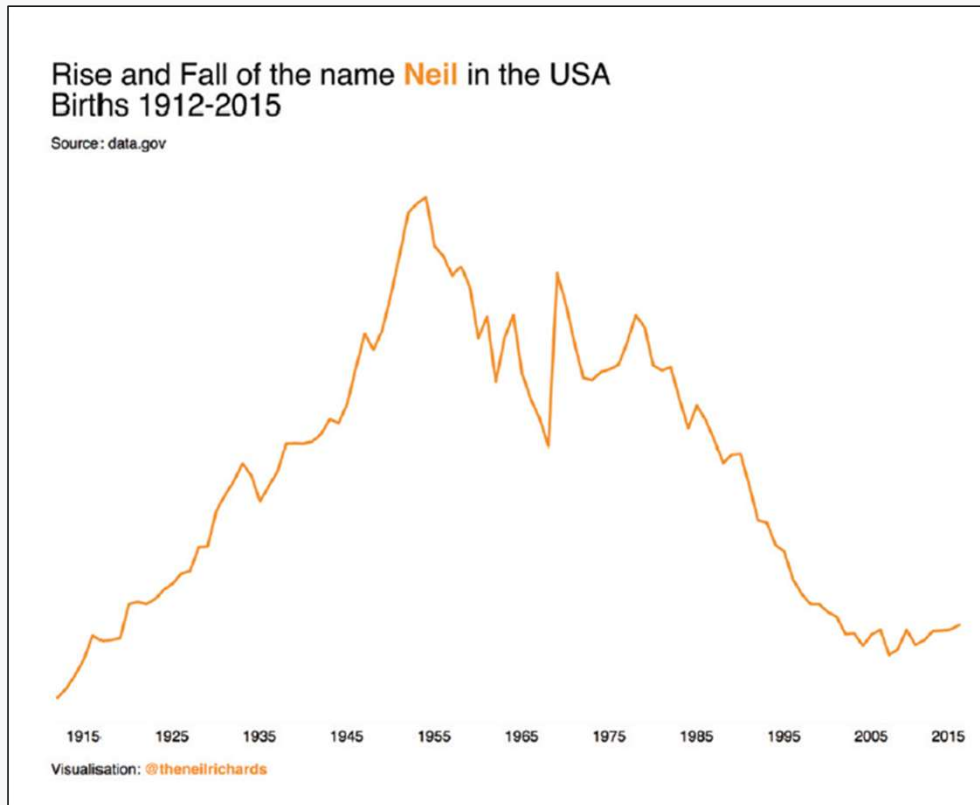
Integrate the graphics and text

Write title like a newspaper headline



Integrate the graphics and the text

Add explainers

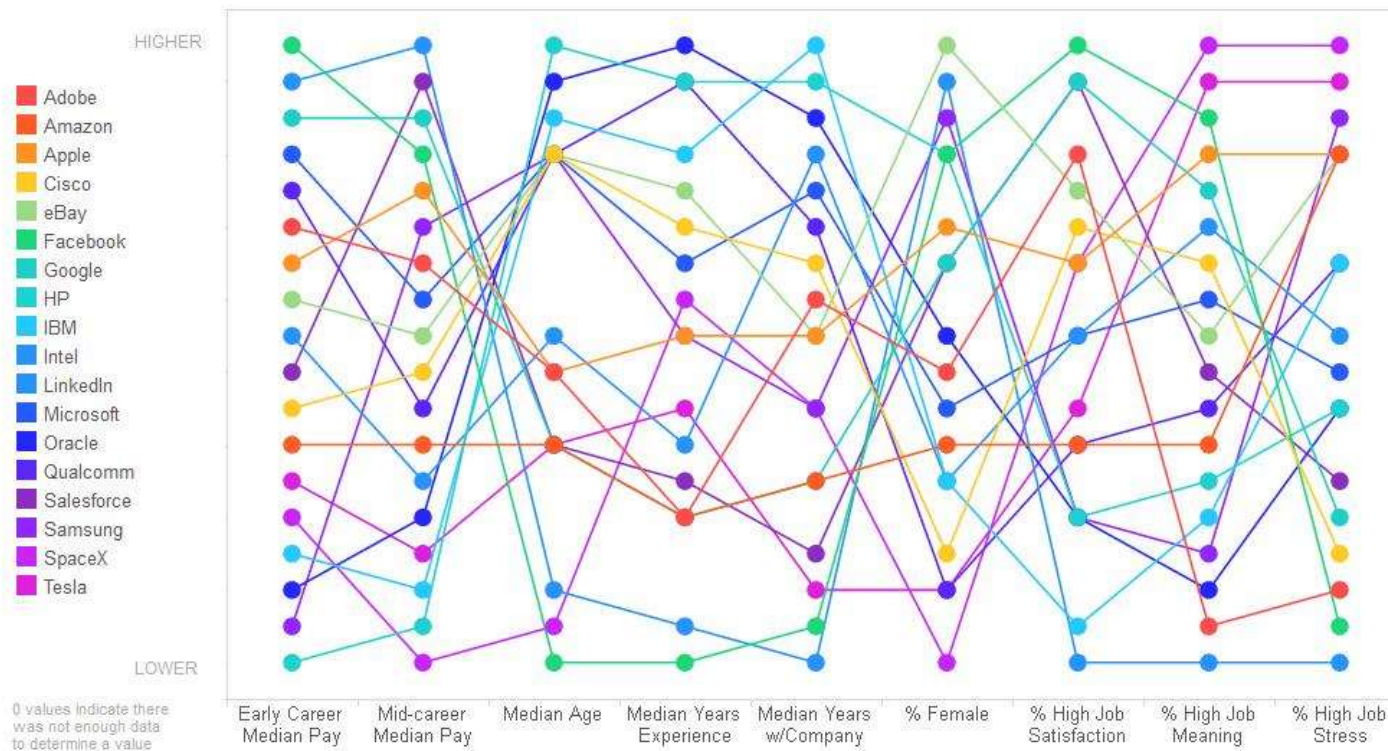


Avoid the “spaghetti chart”

Spot Check: How Do Top Tech Employers Compare?



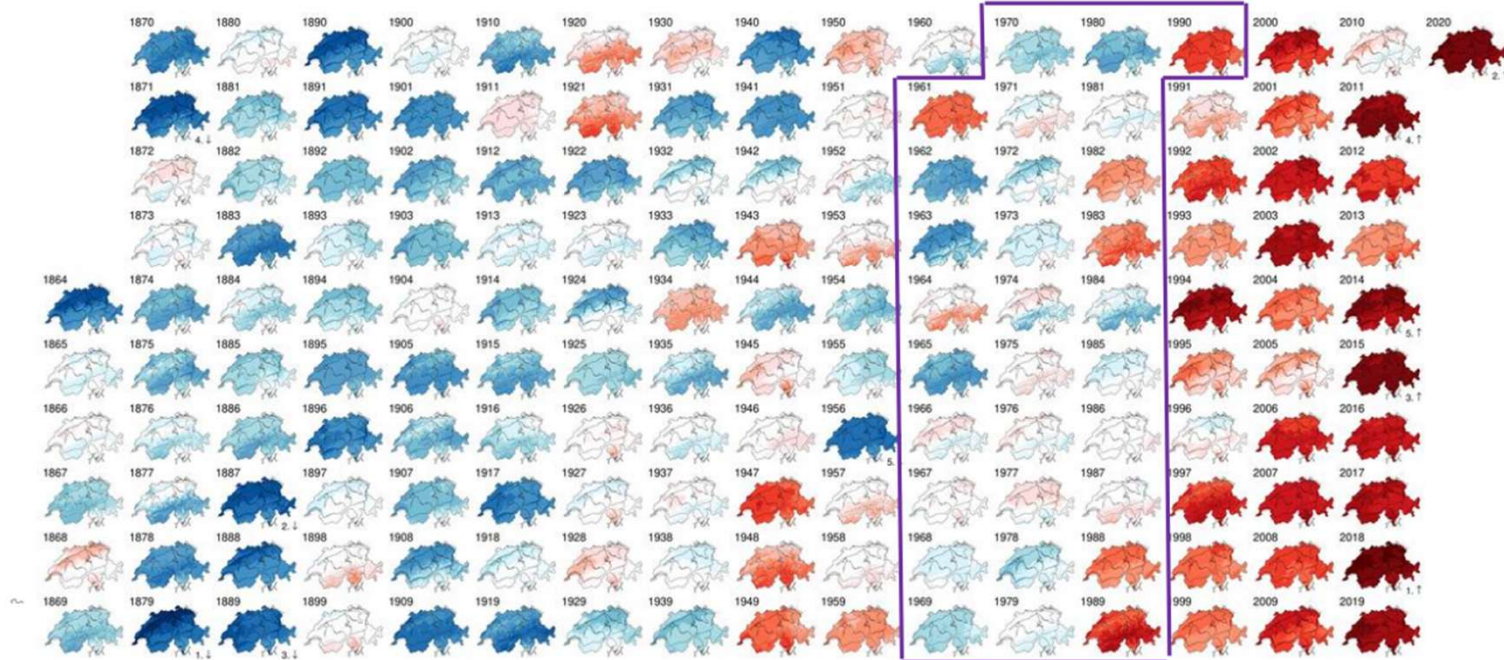
Want to know what it's like to work at a top tech firm like Google, Facebook, or Amazon? PayScale compared 18 tech employers on nine different data points. Which employer looks like the right fit for you?



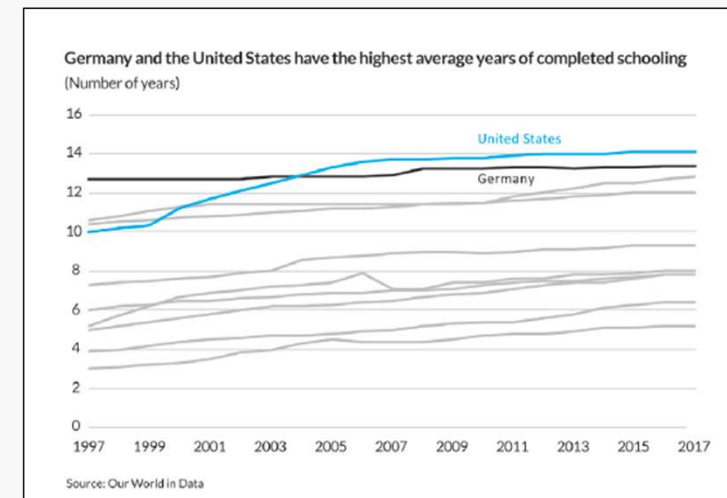
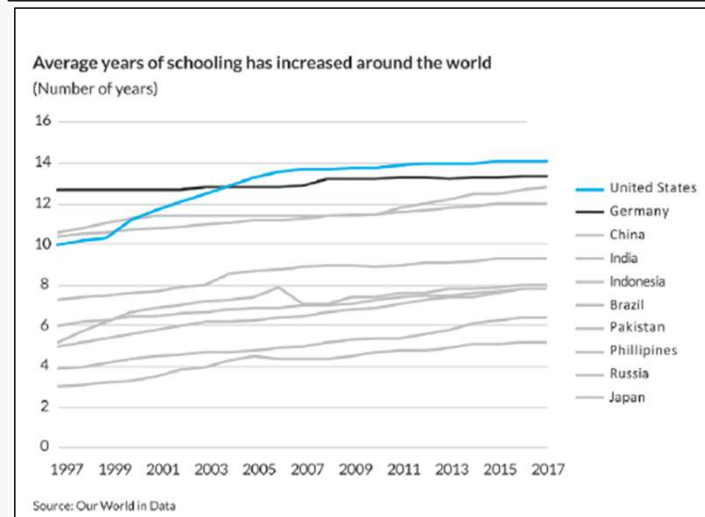
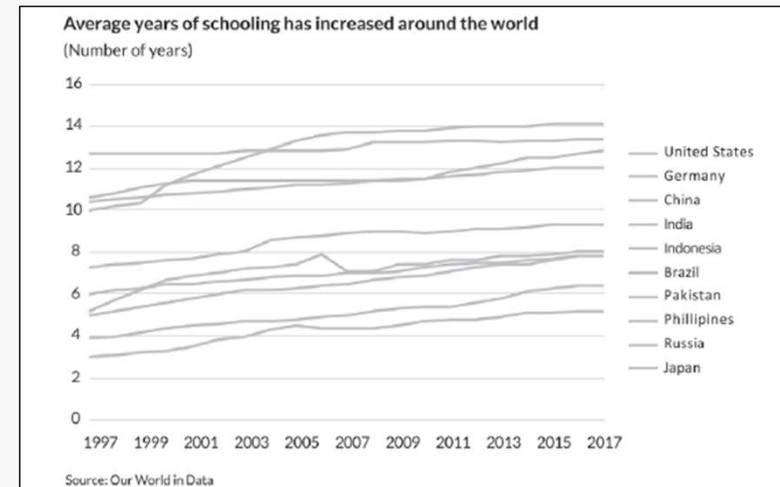
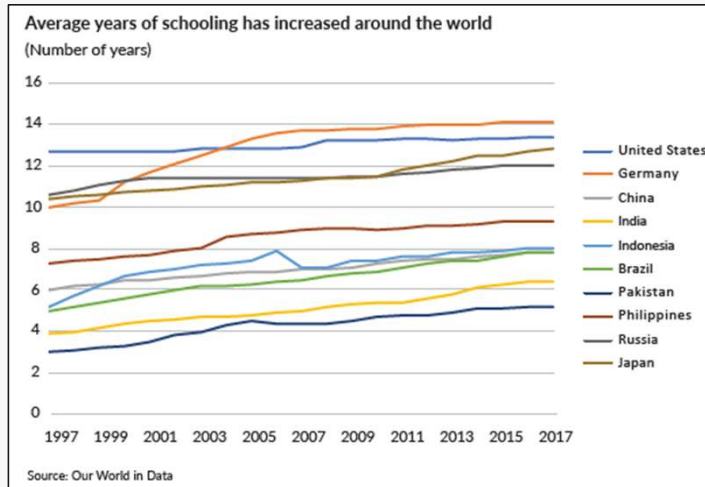
Avoid the “spaghetti chart”

Use grid charts

2010-2019: il decennio più caldo
(dal 1864 / riferimento 1961 - 1990)

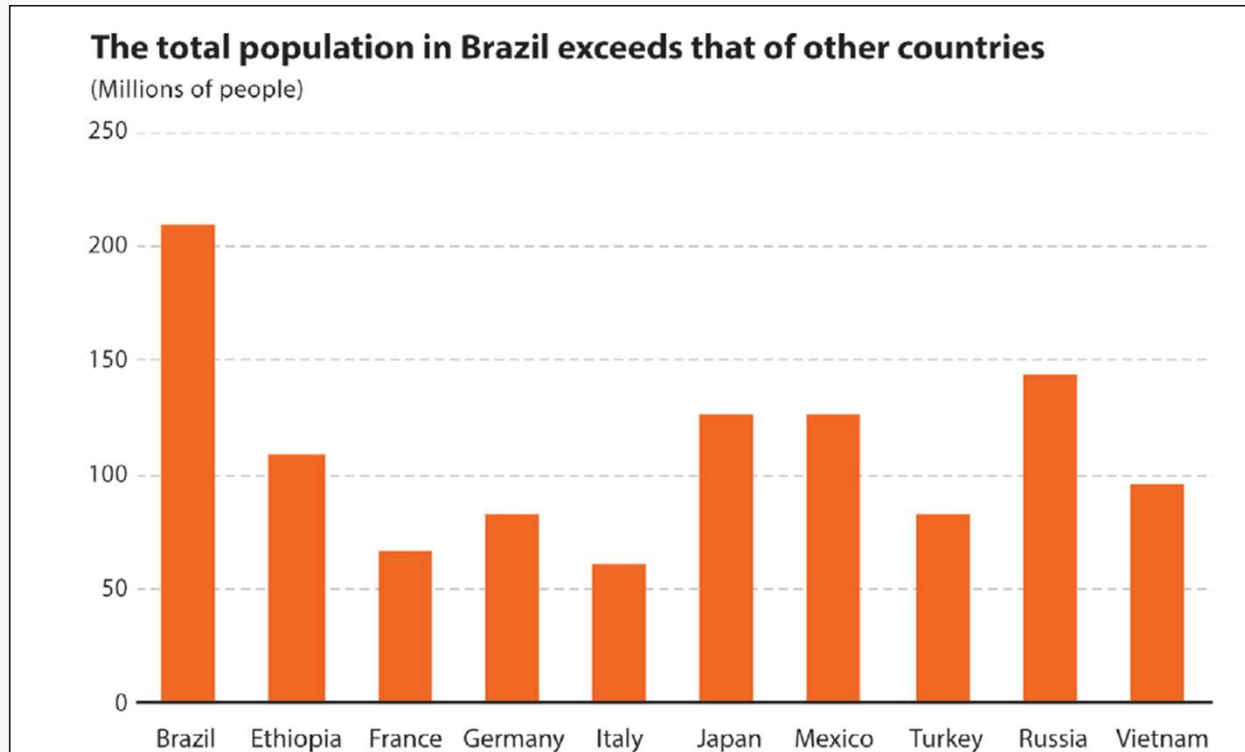


Start with grey



The Bar plot

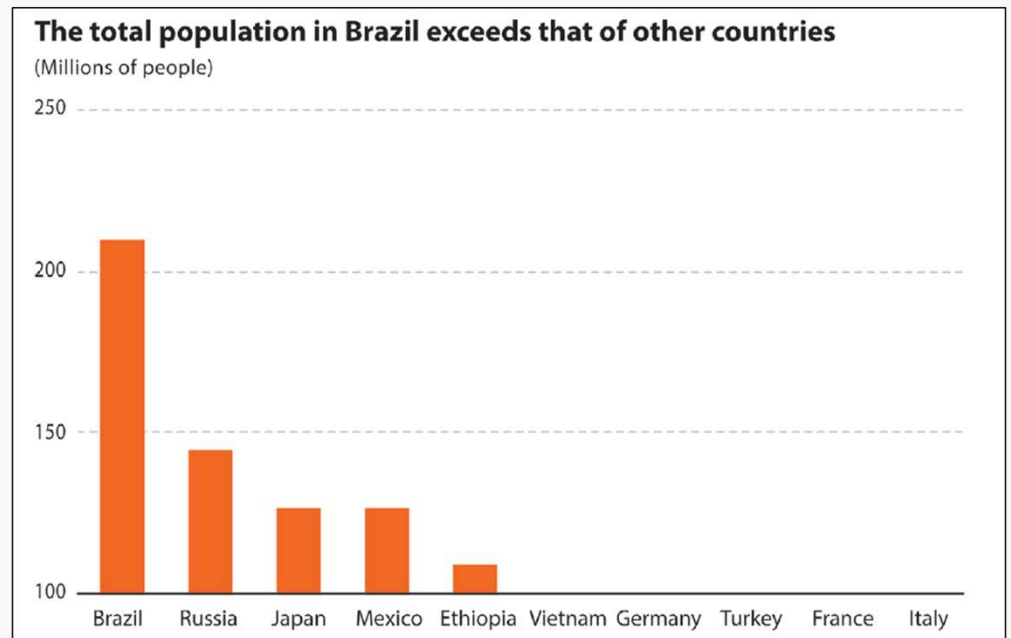
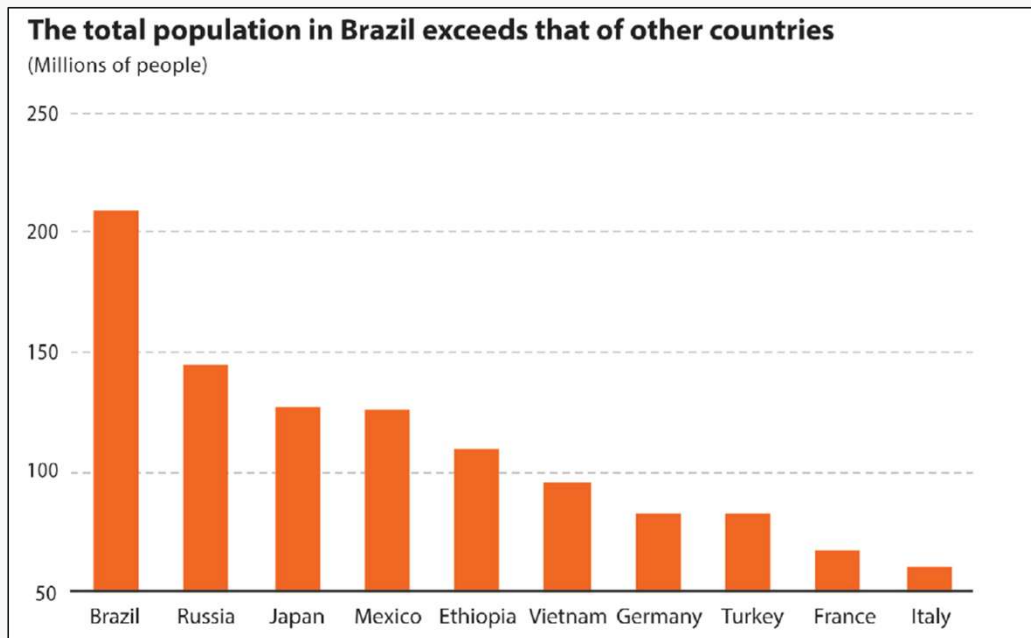
Bar plot



- + Easy to understand
- + Good for comparisons:
- + Versatile
- + Customizable
- + Handles large category sets
- Can become cluttered
- Not ideal for continuous data
- Misleading if not scaled properly
- Takes up space
- Limited trend visualization

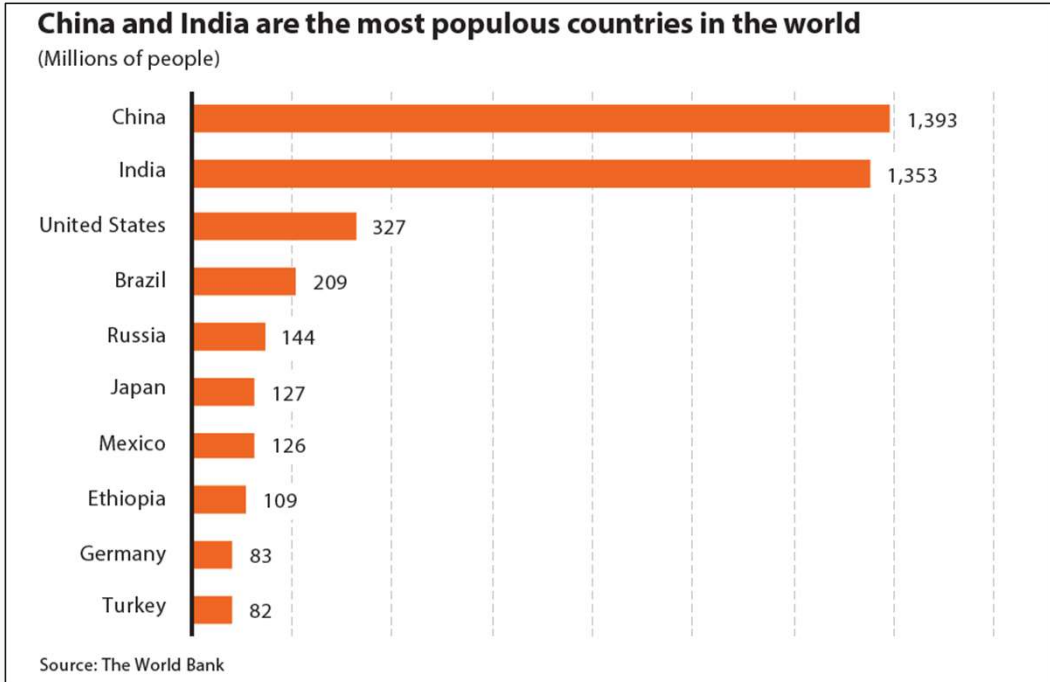
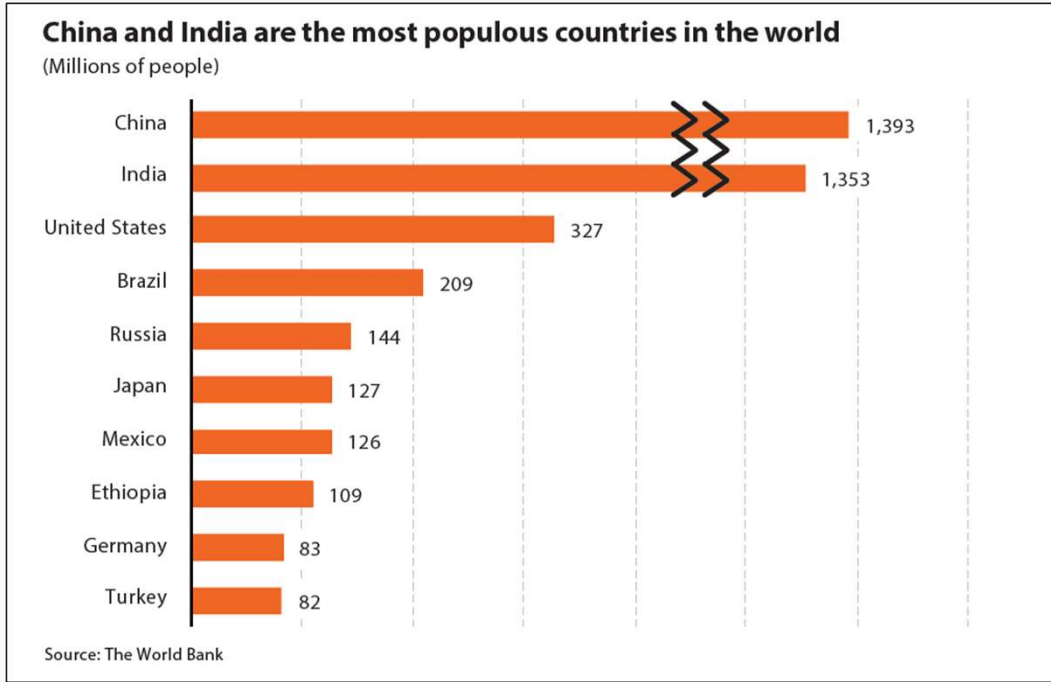
Bar plot

Start the axis at 0



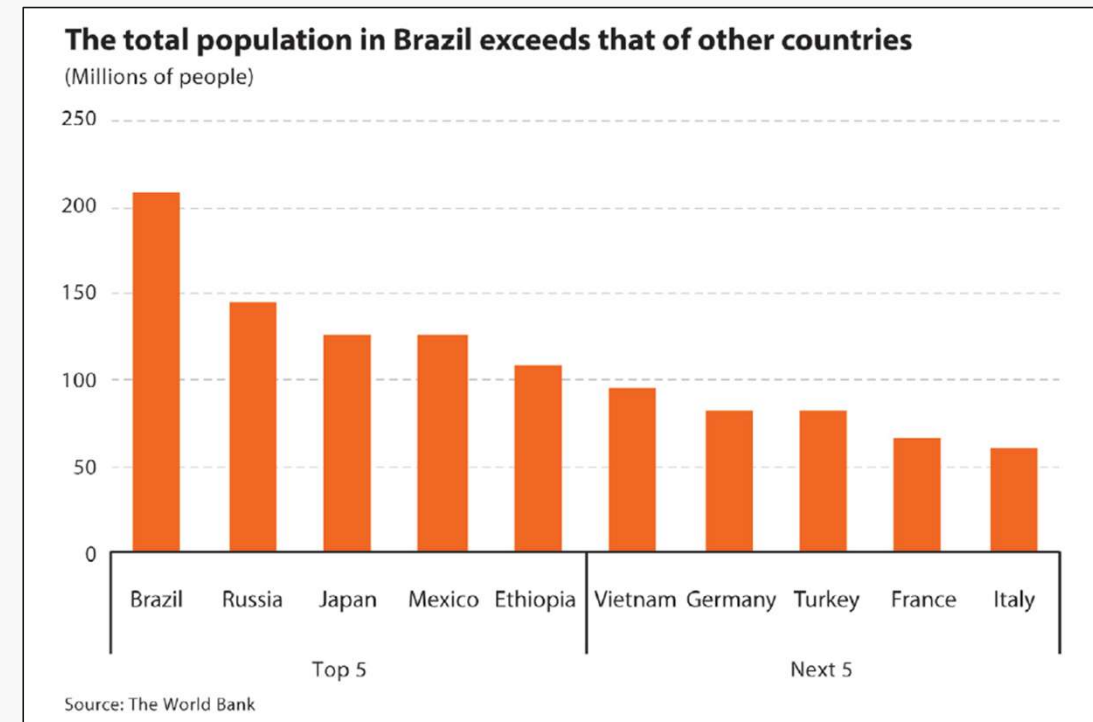
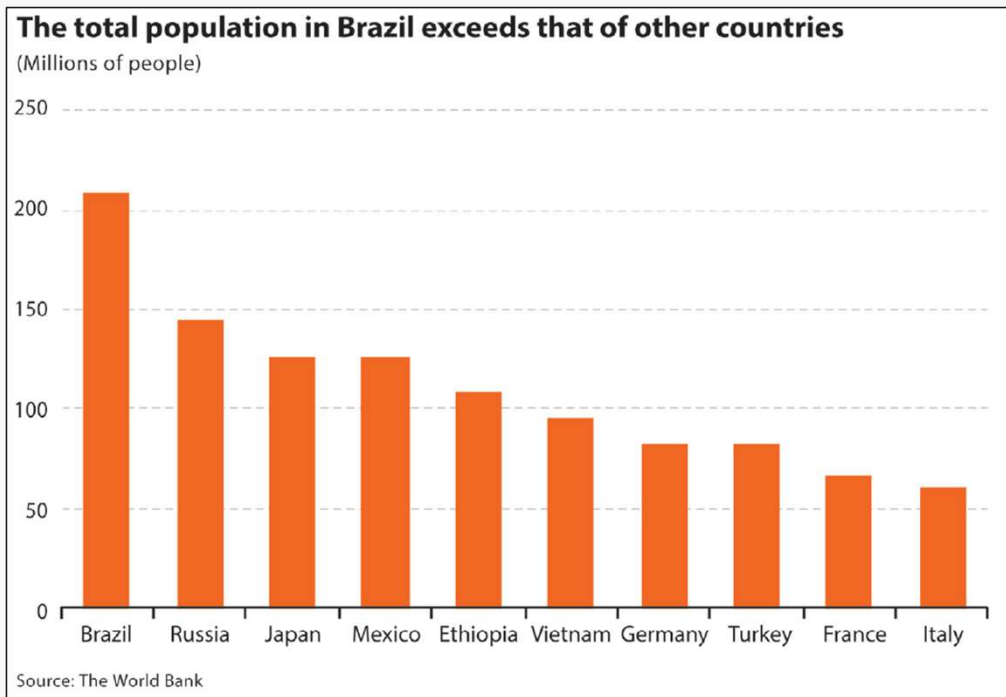
Bar plot

Don't break the bar

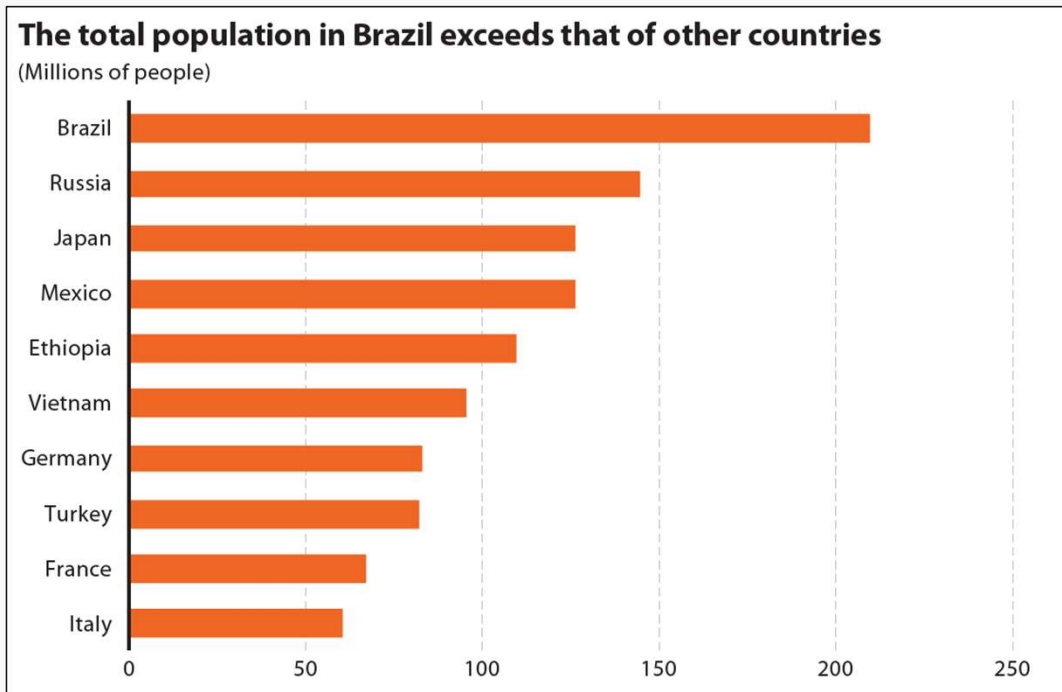


Bar plot

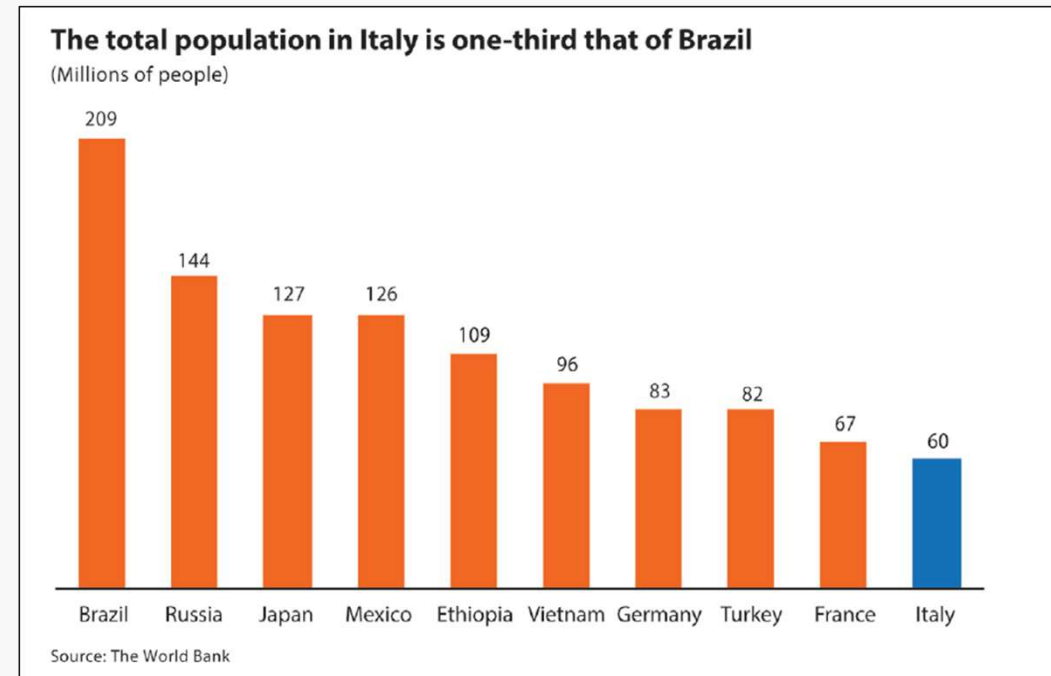
Don't add tick marks for normal bars



Bar plot



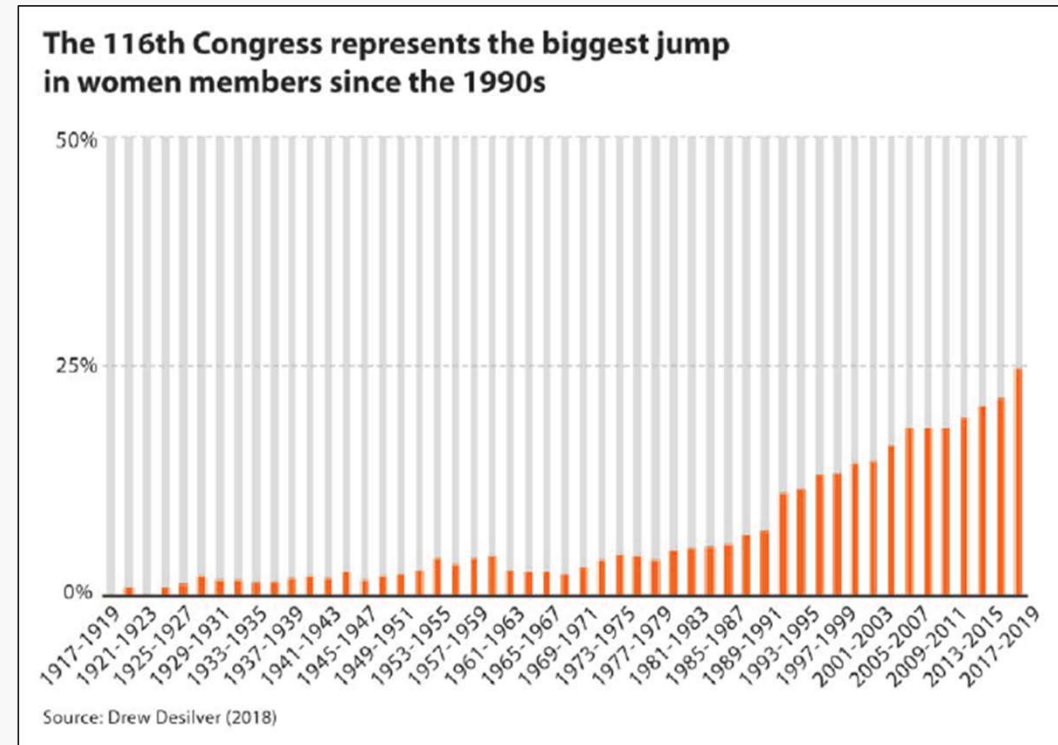
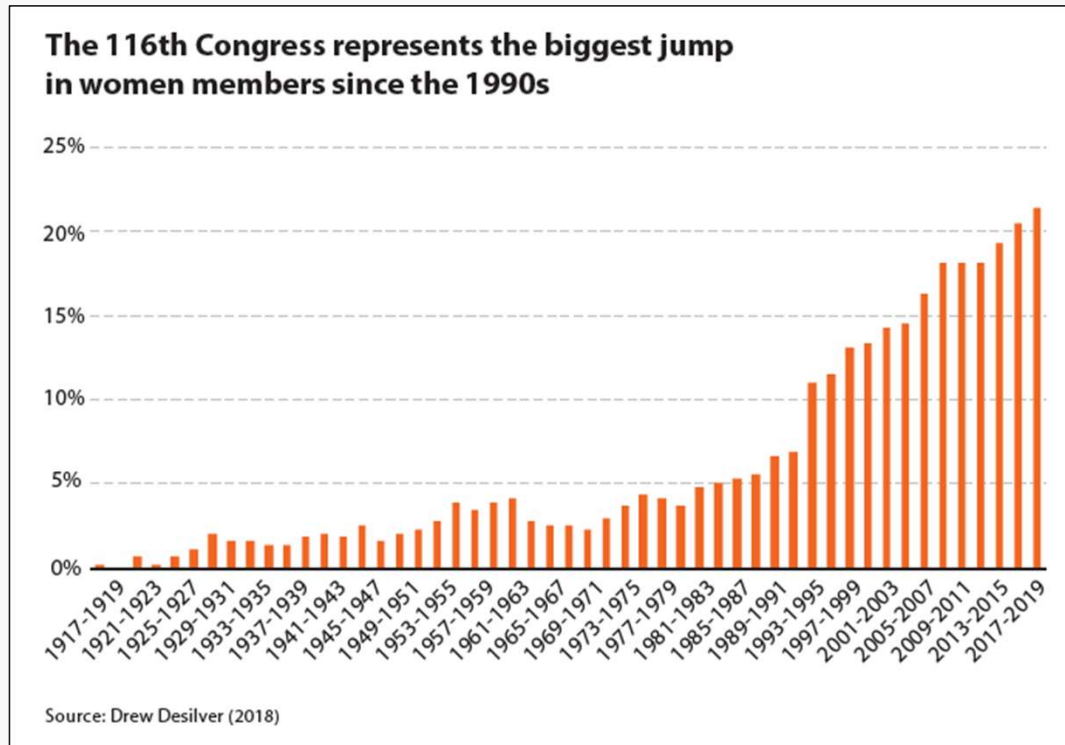
You can rotate the bar for long labels



Data labels make gridline redundant

Bar plot

Show the whole percentage



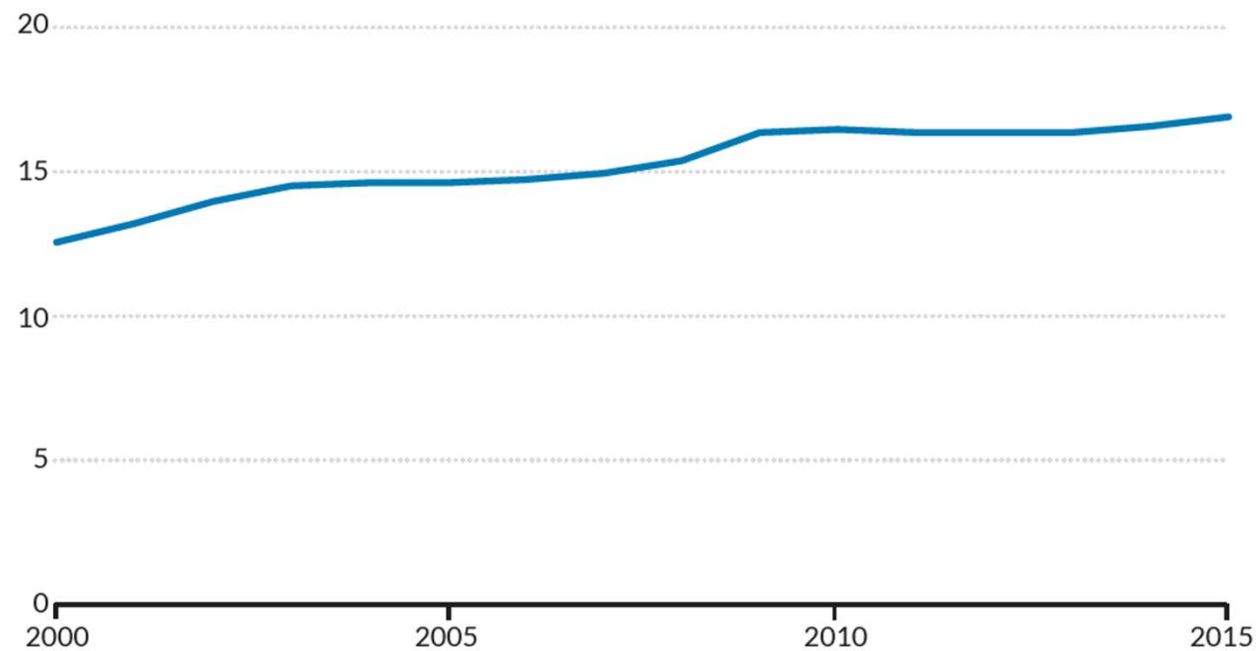
The version on the right shows the same data, but stacks a gray series on top of the data values to 50 percent. In this version, we can emphasize that the share of women is still small even though that share is rising

Time series

Line chart

Total health care spending in the United States grew from 12.5 percent to 16.8 percent between 2000 and 2015

(Percent of GDP)

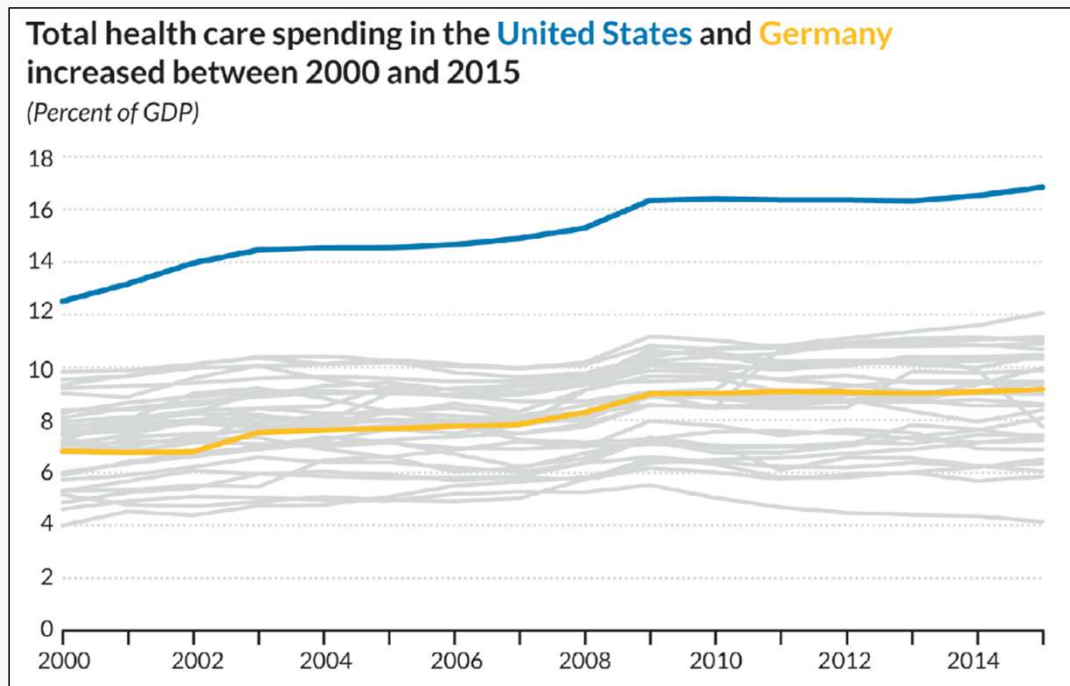


Source: The World Bank

- + Great for showing trends over time
- + Clear data progression
- + Compact and efficient
- + Good for interpolation
- + Highlights patterns
- Not suitable for categorical data
- Overlapping lines can confuse
- Requires consistent intervals
- Data points may be unclear
- Interpolation may mislead

Line chart

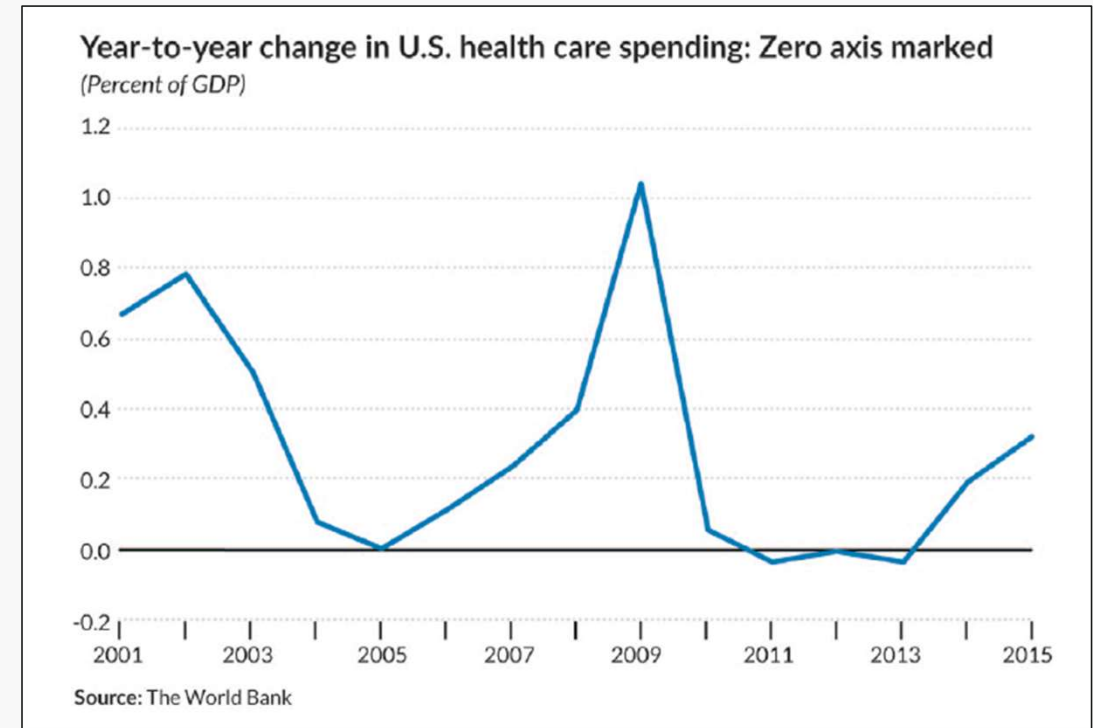
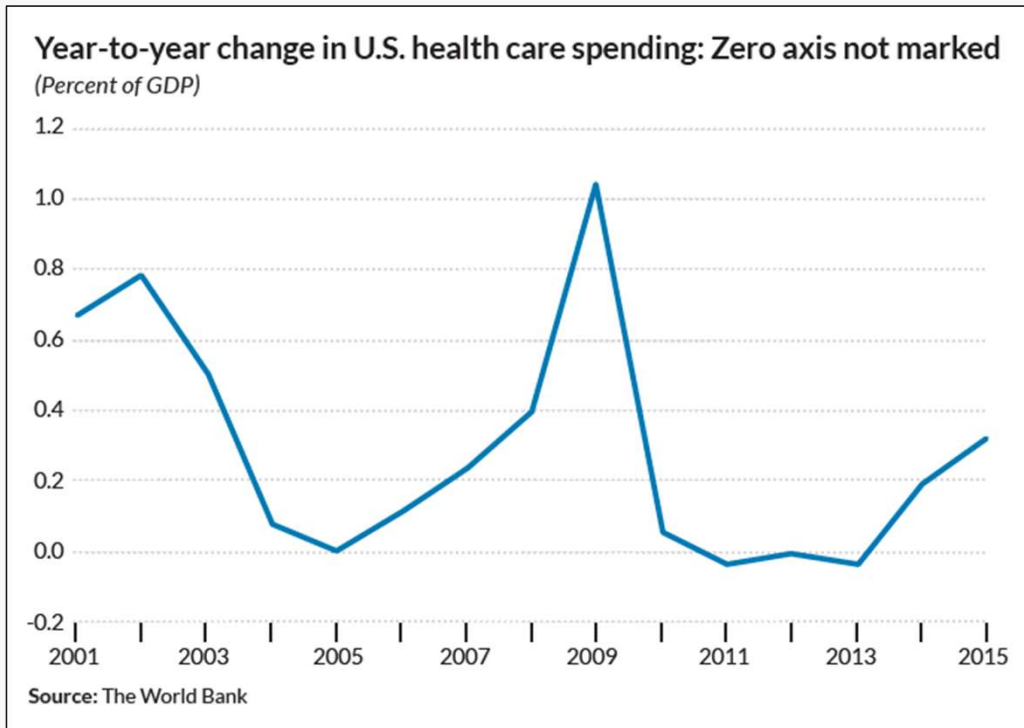
There is no limit to the number of lines you plot



If it gets too messy you can break it up

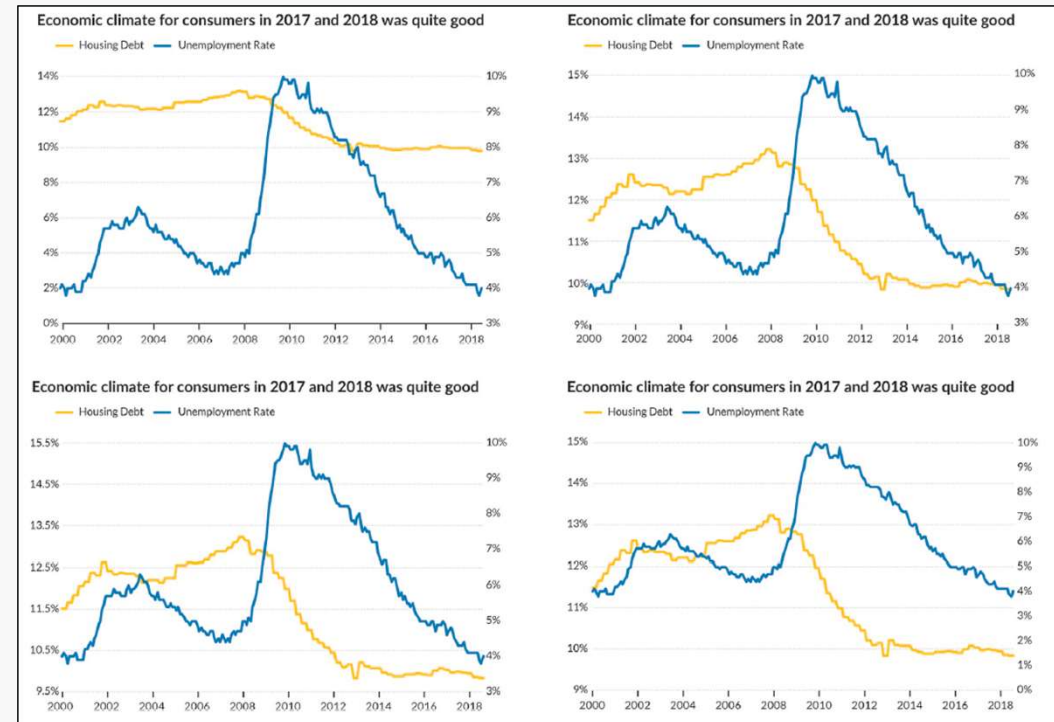
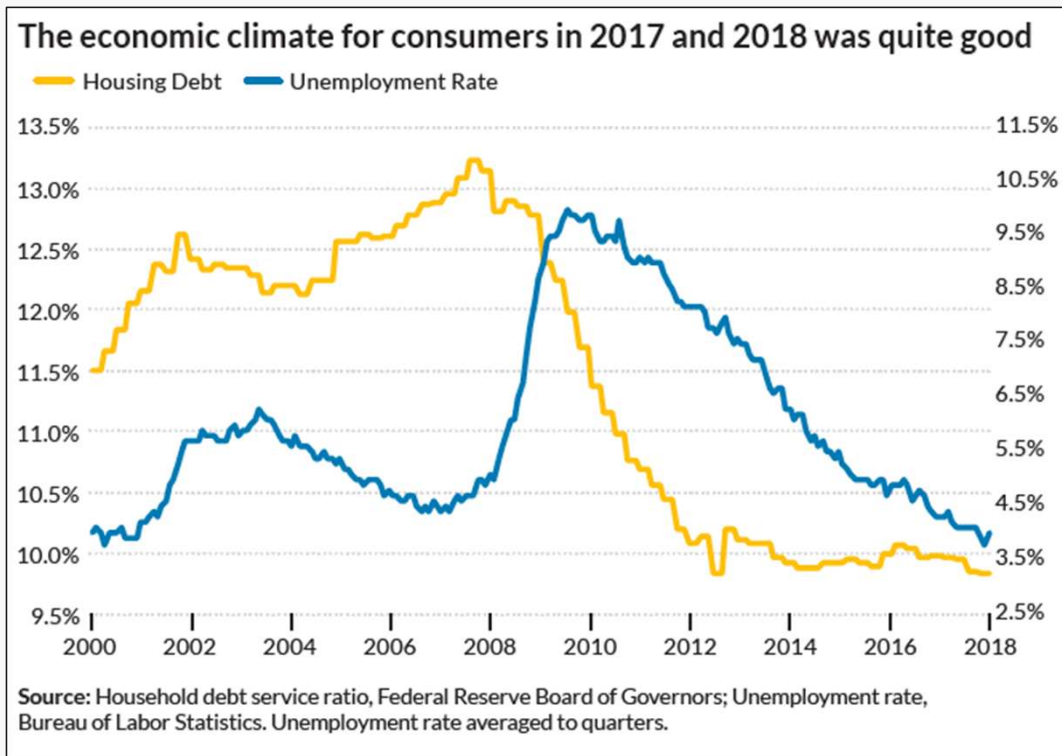
Line chart

We are inclined to think the bottom of the vertical axis is zero



Line chart

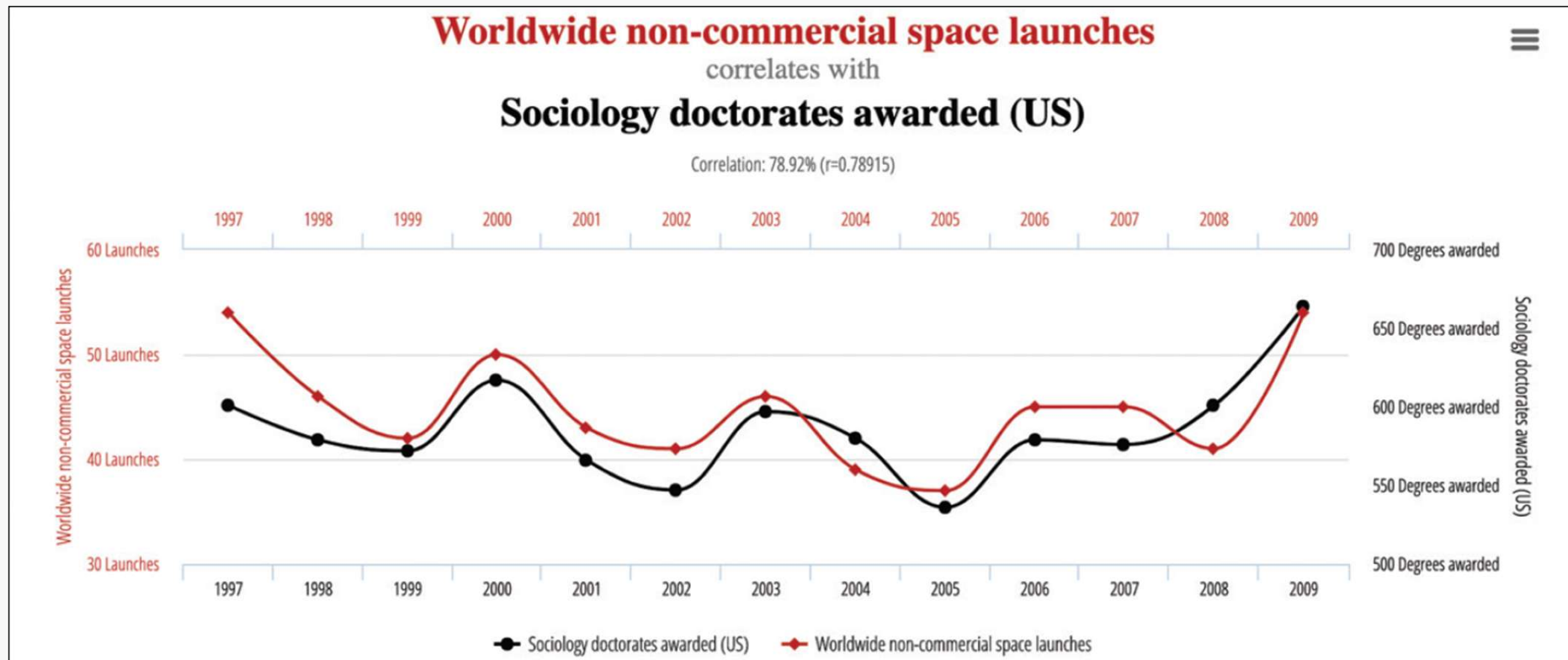
Avoid dual-axis charts



I can choose the axis to distort the meaning

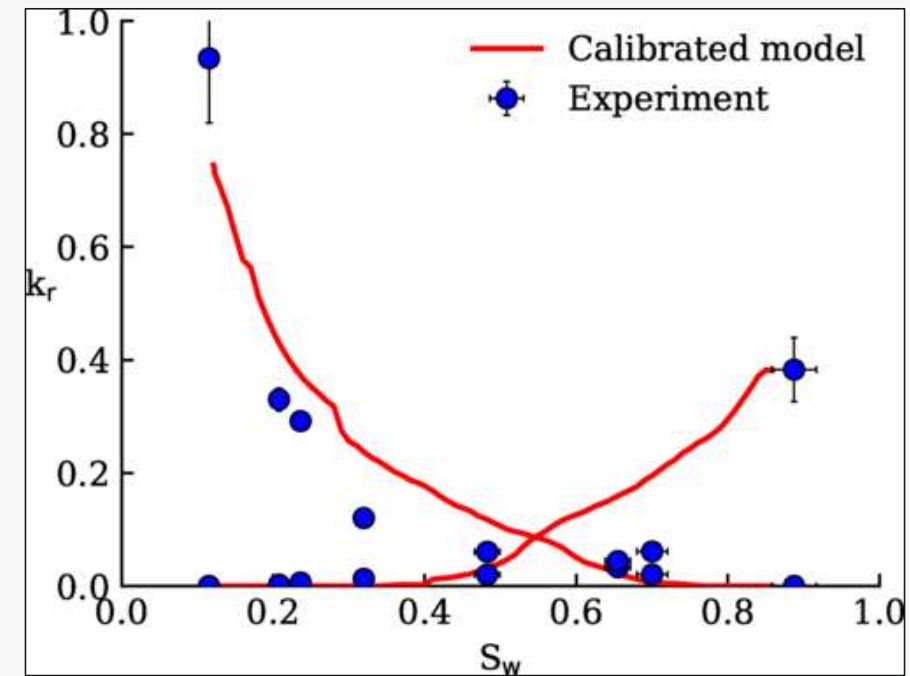
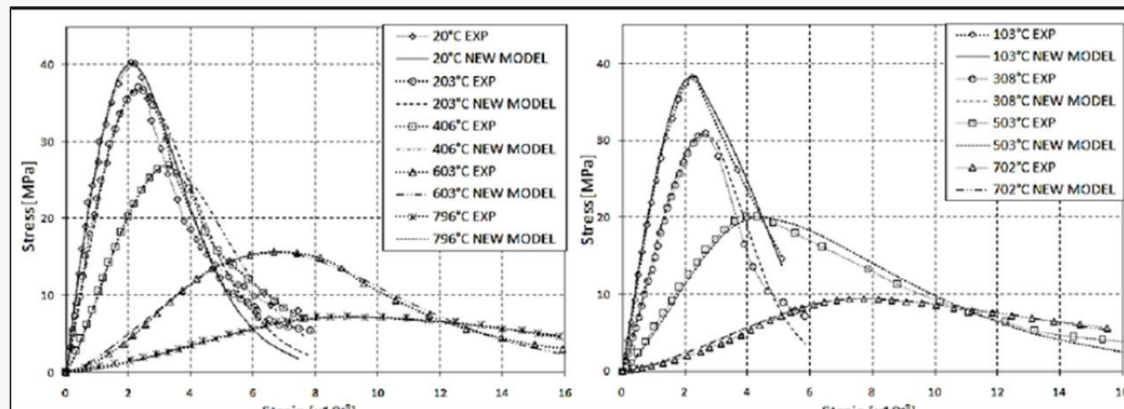
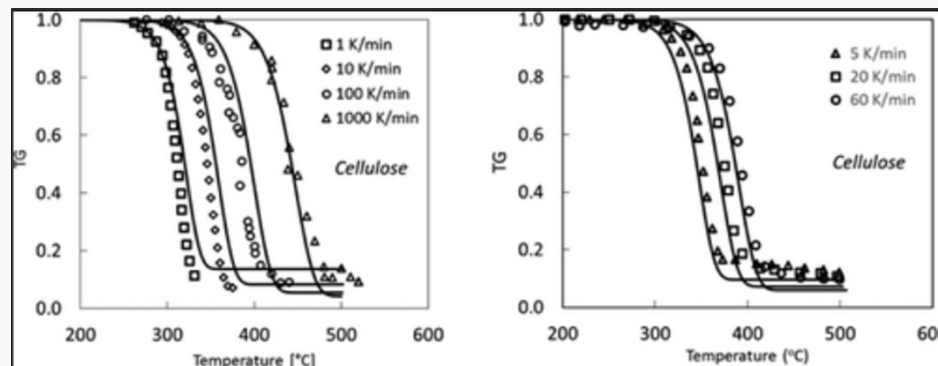
Line chart

You can illude correlations that are non existing



Common practice: experiment and model

Experiments are usually empty indicators, model are whole lines



Valid not only for time series

Tables

Tables

Offset the headers from the body

Country	2013	2014	2015	2016
China	7.23	6.76	6.36	6.12
India	5.10	6.14	6.90	5.89
United States	0.96	1.80	2.09	0.74
Indonesia	4.24	3.73	3.65	3.85
Mexico	-0.06	1.45	1.90	1.68
Pakistan	2.21	2.51	2.61	3.44

Country	2013	2014	2015	2016
China	7.23	6.76	6.36	6.12
India	5.10	6.14	6.90	5.89
United States	0.96	1.80	2.09	0.74
Indonesia	4.24	3.73	3.65	3.85
Mexico	-0.06	1.45	1.90	1.68
Pakistan	2.21	2.51	2.61	3.44

Tables

Use subtle dividers instead of heavy gridlines

Country	2007	2008	2009	2010	2011	Avg.	2012	2013	2014	2015	2016	Avg.
China	13.64	9.09	8.86	10.10	6.36	10.74	7.33	7.23	6.76	6.36	6.12	6.76
India	8.15	2.38	6.95	8.76	6.90	6.30	4.13	5.10	6.14	6.90	5.89	5.63
United States	0.82	-1.23	-3.62	1.68	2.09	-0.30	1.46	0.96	1.80	2.09	0.74	1.41
Indonesia	4.91	4.59	3.24	4.83	3.65	4.47	4.68	4.24	3.73	3.65	3.85	4.03
Mexico	0.70	-0.48	-6.80	3.49	1.90	-0.19	2.15	-0.06	1.45	1.90	1.68	1.41
Pakistan	2.72	-0.36	0.74	-0.48	2.61	0.64	1.34	2.21	2.51	2.61	3.44	2.42
Average	5.15	2.33	1.56	4.73	3.92	3.51	3.52	3.28	3.73	3.92	3.60	3.61

Country	2007	2008	2009	2010	2011	Avg.	2012	2013	2014	2015	2016	Avg.
China	13.64	9.09	8.86	10.10	6.36	10.74	7.33	7.23	6.76	6.36	6.12	6.76
India	8.15	2.38	6.95	8.76	6.90	6.30	4.13	5.10	6.14	6.90	5.89	5.63
United States	0.82	-1.23	-3.62	1.68	2.09	-0.30	1.46	0.96	1.80	2.09	0.74	1.41
Indonesia	4.91	4.59	3.24	4.83	3.65	4.47	4.68	4.24	3.73	3.65	3.85	4.03
Mexico	0.70	-0.48	-6.80	3.49	1.90	-0.19	2.15	-0.06	1.45	1.90	1.68	1.41
Pakistan	2.72	-0.36	0.74	-0.48	2.61	0.64	1.34	2.21	2.51	2.61	3.44	2.42
Average	5.15	2.33	1.56	4.73	3.92	3.51	3.52	3.28	3.73	3.92	3.60	3.61

Tables

Right-align numbers and headers

	2016	2016	2016
China	6,894.40	6,894.40	6,894.40
India	1,862.43	1,862.43	1,862.43
United States	52,319.10	52,319.10	52,319.10
Indonesia	3,974.73	3,974.73	3,974.73
Mexico	9,871.67	9,871.67	9,871.67
Pakistan	1,179.41	1,179.41	1,179.41
Average	12,683.62	12,683.62	12,683.62

Tables

Left align text and header

Right-aligned and hard to read	Centered and even harder to read	Left-aligned and easiest to read
British Virgin Islands Cayman Islands Democratic Republic of Korea Luxembourg United States Germany New Zealand Costa Rica Peru	British Virgin Islands Cayman Islands Democratic Republic of Korea Luxembourg United States Germany New Zealand Costa Rica Peru	British Virgin Islands Cayman Islands Democratic Republic of Korea Luxembourg United States Germany New Zealand Costa Rica Peru

Tables

Select the appropriate level of precision

Country	Too many decimals	Too few decimals	About right
China	6.12380	6	6.1
India	5.88984	6	5.9
United States	0.74279	1	0.7
Indonesia	3.84530	4	3.8
Mexico	1.58236	2	1.6
Pakistan	3.43865	3	3.4
Average	2.63104	3	2.6

Tables

Guide your reader with space between rows and columns

Country	2014	2015	2016
China	6.76	6.36	6.12
India	6.14	6.90	5.89
United States	1.80	2.09	0.74
Indonesia	3.73	3.65	3.85
Mexico	-0.38	-4.37	-4.25
Pakistan	2.51	2.61	3.44
Average	3.43	2.87	2.63

Country	2014	2015	2016
China	6.76	6.36	6.12
India	6.14	6.90	5.89
United States	1.80	2.09	0.74
Indonesia	3.73	3.65	3.85
Mexico	-0.38	-4.37	-4.25
Pakistan	2.51	2.61	3.44
Average	3.43	2.87	2.63

Tables

Group similar data and increase white space.

Region	Country	Per Capita GDP		Percent Change
		2015	2016	
Asia	China	6496.62	6894	6.1238
Asia	India	1758.84	1862	5.8898
North America	United States	51933.40	52319	0.7428
Asia	Indonesia	3827.55	3975	3.8453
North America	Brazil	11351.57	10869	-4.2541
Asia	Pakistan	1140.21	1179	3.4387
Africa	Nigeria	2562.52	2456	-4.1601
Asia	Bangladesh	971.64	1030	5.9627
North America	Mexico	9717.90	9872	1.5824
Asia	Japan	47163.49	47661	1.0546
Africa	Ethiopia	487.29	511	4.9041
Middle East	Egypt	2665.35	2726	2.2633
Europe	Germany	45412.56	45923	1.1240
Middle East	Iran	6007.00	6734	12.1010
Middle East	Turkey	13898.75	14117	1.5734
Europe	France	41642.31	41969	0.7845
Average		15440	15631	2.6860

Region	Country	Per Capita GDP		Percent Change
		2015	2016	
Africa	Ethiopia	487	511	4.90
	Nigeria	2,563	2,456	-4.16
Asia	Bangladesh	972	1,030	5.96
	China	6,497	6,894	6.12
	India	1,759	1,862	5.89
	Indonesia	3,838	3,975	3.85
	Japan	47,163	47,661	1.05
	Pakistan	1,140	1,179	3.44
Europe	France	41,642	41,969	0.78
	Germany	45,413	45,923	1.12
Middle East	Egypt	2,665	2,726	2.26
	Iran	6,007	6,734	12.10
	Turkey	13,899	14,117	1.57
North America	Mexico	9,718	9,872	1.58
	United States	51,933	52,319	0.74
South America	Brazil	11,352	10,869	-4.25
Average		15,440	15,631	2.69

Tables

Add visualizers when appropriate

Country	2007	2016	2007-2016	Country	2016
China	13.64	6.12		China	6.12 
India	8.15	5.89		India	5.89 
United States	0.82	0.74		United States	0.74 
Indonesia	4.91	3.85		Indonesia	3.85 
Mexico	0.70	1.58		Mexico	1.58 
Pakistan	2.72	3.44		Pakistan	3.44 
Average	5.15	3.60		Average	3.60 